

# Mathematical Language

Notation and Conventions

Jungwoo Kim

Department of Mathematics

Pohang University of Science and Technology

## Contents

|                                               |          |
|-----------------------------------------------|----------|
| About This Document                           | 1        |
| <b>Part I. General Mathematical Language</b>  | <b>1</b> |
| 1 General Conventions                         | 1        |
| 2 Mathematical Writing                        | 2        |
| <b>Part II. Notation by Mathematical Area</b> | <b>6</b> |
| 3 Set Theory and Logic                        | 6        |
| 4 Calculus and Analysis                       | 7        |
| 5 Linear Algebra                              | 12       |
| 6 Differential Equations                      | 14       |
| 7 Combinatorics and Discrete Mathematics      | 16       |
| 8 Probability and Statistics                  | 17       |
| 9 Geometry                                    | 19       |
| 10 Number Theory                              | 21       |
| 11 Algebra                                    | 22       |
| 12 Topology                                   | 24       |
| 13 Miscellaneous Notation                     | 25       |

## About This Document

Mathematical Language records the notation and writing conventions adopted here. It is guided by deliberate personal preferences together with established mathematical practice; when no personal convention has been fixed, standard mathematical usage is the default.

This document is intentionally non-exhaustive and may be extended as new subjects and notation are encountered. A displayed example or notation-table entry gives a standard or preferred form, not an exhaustive grammar for every related mathematical expression, unless explicitly stated otherwise. The absence of a notation from this document does not by itself make that notation invalid.

When notation not covered here is needed, an established standard should be used when suitable; otherwise, a clear, mathematically coherent, and unambiguous notation appropriate to the context may be introduced. Useful choices may later be incorporated into this document. These conventions should be applied with mathematical judgment rather than mechanically: explicitly stated conventions should be respected, while established field-specific usage, semantic distinctions, and context take precedence over superficial uniformity.

## Part I. General Mathematical Language

### 1 General Conventions

Parentheses are preferred when delimiters are otherwise interchangeable, even in nested expressions; nesting alone is not a reason to alternate among parentheses, square brackets, and braces. Widely used standard alternatives may be listed in preferred order; established field-specific or semantically distinct delimiters override this preference.

| Item                       | Convention                                                                                                                                                                                                                                                             |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scalars                    | Ordinary mathematical italic letters, for example $a, b, c, \lambda, t$ .                                                                                                                                                                                              |
| Vectors                    | Bold letters; generic vectors use lowercase letters such as $\mathbf{u}, \mathbf{v}, \mathbf{x}$ , while established field-specific vector names may use other letter cases. Unless otherwise stated, vectors are column vectors.                                      |
| Matrices                   | Bold uppercase letters, for example $\mathbf{A}, \mathbf{B}, \mathbf{M}$ . Parentheses are the preferred matrix delimiters; square brackets are a standard alternative.                                                                                                |
| Greek vectors and matrices | Bold Greek letters when a Greek symbol denotes a vector or matrix, for example $\boldsymbol{\xi}$ and $\boldsymbol{\Phi}$ .                                                                                                                                            |
| Dimension subscripts       | Prefer dimension subscripts when the dimension is readily available and useful; omit them when it is clear, immaterial, or not readily available rather than inferring a dimension solely to supply a subscript.                                                       |
| Number systems             | Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ .                                                                                                                                                                            |
| Fields                     | Arbitrary or named fields are denoted by ordinary italic uppercase letters such as $F, K, L$ . Standard finite fields such as $\mathbb{F}_q$ retain blackboard bold; other established blackboard-bold notation follows the standard convention of the relevant field. |
| Sets                       | Ordinary uppercase letters such as $A, B, X, Y$ , unless a field-specific convention requires otherwise.                                                                                                                                                               |
| Families and collections   | Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$ .                                                                                                                                                                                |
| Named operators            | Upright Roman type, for example $\det, \ker, \operatorname{im}, \operatorname{rk}, \operatorname{tr}$ .                                                                                                                                                                |
| Large operators            | Display size in displayed mathematics and notation tables; sums and products place indices above and below, limits place subscripts below, and integral bounds retain standard right-side placement.                                                                   |

|                |                                                                                                                                                                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Greek variants | Distinct typographic variants of Greek letters are treated as distinct symbols, not as interchangeable forms of the same symbol. For example, $\varepsilon$ and $\epsilon$ , and $\varphi$ and $\phi$ , are distinct unless explicitly identified otherwise. |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

| Notation                    | Meaning                                                                         |
|-----------------------------|---------------------------------------------------------------------------------|
| $(a_1, \dots, a_n)$         | Ordered tuple.                                                                  |
| $(a_n), (a_n)_{n \geq 1}$   | Sequence; an appropriate natural-number index set may be displayed when useful. |
| $(a_i)_{i \in I}$           | Indexed family over an arbitrary index set $I$ .                                |
| $(a, b)$                    | Open interval.                                                                  |
| $[a, b]$                    | Closed interval.                                                                |
| $[a, b)$                    | Left-closed, right-open interval.                                               |
| $(a, b]$                    | Left-open, right-closed interval.                                               |
| $\{a_1, \dots, a_n\}$       | Set delimiters.                                                                 |
| $\{x \in A \mid P(x)\}$     | Set-builder notation.                                                           |
| $\langle x, y \rangle$      | Pairing or inner product, according to context.                                 |
| $ x $                       | Absolute value, modulus, cardinality, or order, according to context.           |
| $\ x\ $                     | Norm.                                                                           |
| s.t.                        | Abbreviation for “such that” when naturally required.                           |
| $a = b$                     | Equality.                                                                       |
| $a := b$                    | $a$ is defined to be $b$ .                                                      |
| $a \equiv b$                | Identity, equivalence, or congruence, according to context.                     |
| $a \cong b$                 | Isomorphism or congruence, according to context.                                |
| $a \approx b$               | Approximation.                                                                  |
| $f: X \rightarrow Y$        | Function or map from $X$ to $Y$ .                                               |
| $x \mapsto f(x)$            | Elementwise rule of a map.                                                      |
| $g \circ f$                 | Composition: first $f$ , then $g$ .                                             |
| $f _A, f \upharpoonright_A$ | Restriction of $f$ to $A$ .                                                     |
| $\text{id}_X$               | Identity map on $X$ .                                                           |

## 2 Mathematical Writing

### Mathematical Environments.

| Item        | Convention                                                                                                                                                     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition  | Precise meaning of a term, object, property, or construction.                                                                                                  |
| Notation    | Introduction or declaration of notation.                                                                                                                       |
| Convention  | Explicit choice of notation, terminology, normalization, or usage adopted in the stated context.                                                               |
| Axiom       | Fundamental statement accepted without proof as part of a mathematical system or theory.                                                                       |
| Assumption  | Condition adopted as a premise for the current statement, argument, construction, or discussion.                                                               |
| Hypothesis  | Distinguished condition or posited statement under which a result or theory is considered; established named hypotheses retain their conventional terminology. |
| Conjecture  | Statement proposed to be true but not proved in full generality.                                                                                               |
| Theorem     | Principal mathematical result.                                                                                                                                 |
| Proposition | Useful result of more limited scope.                                                                                                                           |

| Item           | Convention                                                                                                          |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| Lemma          | Supporting result, usually for a later proof.                                                                       |
| Corollary      | Result following readily from a preceding result.                                                                   |
| Claim          | Local or intermediate assertion within a larger argument.                                                           |
| Fact           | Established statement used without the emphasis or status of a named theorem.                                       |
| Observation    | Short mathematical insight or consequence that follows readily from the context, definitions, or preceding results. |
| Proof          | Rigorous argument establishing a stated mathematical result.                                                        |
| Sketch         | Outline of the essential argument, construction, or solution with some details omitted.                             |
| Formula        | Distinguished computational identity or closed form.                                                                |
| Algorithm      | Explicit step-by-step procedure.                                                                                    |
| Example        | Illustration of a definition, construction, phenomenon, or result.                                                  |
| Counterexample | Example disproving a general assertion or showing a hypothesis is necessary.                                        |
| Remark         | Supplementary mathematical observation, clarification, qualification, or consequence.                               |
| Comment        | Expository or editorial note concerning interpretation, notation, presentation, or usage.                           |
| Warning        | Important notice of a subtlety, common error, or potentially misleading point.                                      |
| Problem        | Mathematical task intended to be solved.                                                                            |
| Exercise       | Task primarily intended to practice, apply, or reinforce mathematical material.                                     |
| Open Problem   | Problem not known to be solved in the stated form.                                                                  |
| Solution       | Resolution of a stated problem or task by proof, computation, construction, or other appropriate means.             |
| Idea           | Decisive conceptual insight.                                                                                        |
| Strategy       | Broad problem-solving plan.                                                                                         |
| Tactic         | Local step within a larger strategy.                                                                                |

Proof, sketch, and solution labels are written as *Proof*, *Sketch*, and *Solution*, respectively, each followed by a period.

| Notation | Meaning                                                                 |
|----------|-------------------------------------------------------------------------|
| □        | Ending for ordinary proofs and solutions.                               |
| ■        | Ending for substantial, difficult, or particularly important proofs.    |
| Q.E.D.   | Ending for exceptionally difficult, significant, or paper-level proofs. |

**Enumeration and Subdivision.** Enumeration labels are set in upright Roman type. Their form is determined by structural role rather than nesting depth.

| Item               | Convention                                                                                                                                                                                               |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1, 1.2, 1.3, ... | Independent child problems or standalone units under a common parent; each is a distinct subproblem that may be stated, solved, and referenced separately.                                               |
| (1), (2), (3), ... | Ordinary parallel enumeration, such as choices, cases, possibilities, alternatives, examples, or procedural steps; not a subdivision of a single problem or a clause of a formal mathematical statement. |

| Item                  | Convention                                                                                                                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a), (b), (c), ...    | Connected subparts of a single problem or task that share a common statement, setup, or context.                                                                                           |
| (i), (ii), (iii), ... | Formal mathematical clauses within definitions, theorems, propositions, lemmas, and related statements, such as hypotheses, conditions, properties, conclusions, or equivalent statements. |

The distinction between 1.1 and (a) is independence: 1.1 denotes a distinct child problem under a common parent, whereas (a) denotes a connected part of one problem sharing its setup. The distinction between (1) and (i) is function: (1) is used for ordinary parallel enumeration, whereas (i) is reserved for clauses within formal mathematical statements. Nested enumeration preserves these structural roles rather than following a fixed hierarchy of label styles.

### Mathematical Shorthand.

Typeset abbreviations may be used in typeset mathematical writing as well as handwritten work. Handwritten shorthand is reserved for notes, scratch work, and blackboard writing.

#### *Typeset Abbreviations.*

| Notation | Meaning                                                                                 |
|----------|-----------------------------------------------------------------------------------------|
| i.e.     | “that is.”                                                                              |
| i.o.w.   | “in other words.”                                                                       |
| s.t.     | “such that”; preferred when an abbreviation is natural in typeset mathematical writing. |
| iff      | “if and only if.”                                                                       |
| w.r.t.   | “with respect to.”                                                                      |
| const    | “constant.”                                                                             |
| N.t.     | “note that.”                                                                            |
| w.l.o.g. | “without loss of generality.”                                                           |
| w.t.s.   | “want to show (that).”                                                                  |
| e.t.s.   | “enough to show (that).”                                                                |
| t.f.a.e. | “the following are equivalent.”                                                         |
| i.t.s.w. | “in the same way.”                                                                      |
| o.t.o.h. | “on the other hand.”                                                                    |
| IVT      | Intermediate Value Theorem.                                                             |
| EVT      | Extreme Value Theorem.                                                                  |
| MVT      | Mean Value Theorem.                                                                     |
| FTC      | Fundamental Theorem of Calculus.                                                        |
| ODE      | Ordinary Differential Equation.                                                         |
| PDE      | Partial Differential Equation.                                                          |
| IVP      | Initial Value Problem.                                                                  |
| BVP      | Boundary Value Problem.                                                                 |

#### *Handwritten Shorthand.*

| Notation | Meaning        |
|----------|----------------|
| Def      | “definition.”  |
| Thm      | “theorem.”     |
| Prop     | “proposition.” |
| Lem      | “lemma.”       |
| Cor      | “corollary.”   |

| Notation | Meaning             |
|----------|---------------------|
| Pf       | “proof.”            |
| Sol      | “solution.”         |
| Ex       | “example.”          |
| Exer     | “exercise.”         |
| Rmk      | “remark.”           |
| Sps      | “suppose (that).”   |
| cont     | “continuous.”       |
| cont.ly  | “continuously.”     |
| difb     | “differentiable.”   |
| difb.ly  | “differentiably.”   |
| difl     | “differential.”     |
| bdd      | “bounded.”          |
| unif     | “uniform.”          |
| unif.ly  | “uniformly.”        |
| lin      | “linear.”           |
| lin.ly   | “linearly.”         |
| dep      | “dependent.”        |
| indep    | “independent.”      |
| approx   | “approximate.”      |
| triv     | “trivial.”          |
| fnc      | “function.”         |
| seq      | “sequence.”         |
| eqn      | “equation.”         |
| eqlt     | “equality.”         |
| ineqlt   | “inequality.”       |
| sys      | “system.”           |
| comb     | “combination.”      |
| vec      | “vector.”           |
| mat      | “matrix.”           |
| col      | “column.”           |
| pt       | “point.”            |
| bdy      | “boundary.”         |
| nbhd     | “neighborhood.”     |
| itvl     | “interval.”         |
| op       | “operation.”        |
| num      | “number.”           |
| val      | “value.”            |
| elmt     | “element.”          |
| cw       | “clockwise.”        |
| ccw      | “counterclockwise.” |
| w/       | “with.”             |
| w/o      | “without.”          |

**Mathematical Phrases.**

| Phrase                         | Use                                                      |
|--------------------------------|----------------------------------------------------------|
| We find by inspection that ... | Used when a conclusion follows directly from inspection. |

**Part II. Notation by Mathematical Area****3 Set Theory and Logic**

Uppercase letters usually denote sets; lowercase letters usually denote their elements.

| Notation                               | Meaning                                                                                            |
|----------------------------------------|----------------------------------------------------------------------------------------------------|
| $\mathbb{N} = \{1, 2, 3, \dots\}$      | Positive integers.                                                                                 |
| $\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$ | Nonnegative integers.                                                                              |
| $\mathbb{Z}$                           | Integers.                                                                                          |
| $\mathbb{Q}$                           | Rational numbers.                                                                                  |
| $\mathbb{R}$                           | Real numbers.                                                                                      |
| $\mathbb{C}$                           | Complex numbers.                                                                                   |
| $\emptyset, \varnothing$               | Empty set.                                                                                         |
| $x \in A$                              | $x$ is an element of $A$ .                                                                         |
| $x \notin A$                           | $x$ is not an element of $A$ .                                                                     |
| $A \subseteq B$                        | $A$ is a subset of $B$ ; equality is allowed.                                                      |
| $A \subsetneq B$                       | $A$ is a proper subset of $B$ .                                                                    |
| $A \supseteq B$                        | $A$ is a superset of $B$ ; equality is allowed.                                                    |
| $A \supsetneq B$                       | $A$ is a proper superset of $B$ .                                                                  |
| $A \cup B$                             | Union.                                                                                             |
| $A \cap B$                             | Intersection.                                                                                      |
| $(A_i)_{i \in I}$                      | Indexed family of sets.                                                                            |
| $\bigcup_{i \in I} A_i$                | Union of an indexed family of sets.                                                                |
| $\bigcap_{i \in I} A_i$                | Intersection of an indexed family of sets.                                                         |
| $A \setminus B, A - B$                 | Set difference.                                                                                    |
| $A \triangle B, A \Delta B$            | Symmetric difference; $A \triangle B$ is preferred.                                                |
| $A^c$                                  | Complement of $A$ in the ambient set.                                                              |
| $\mathcal{P}(A), 2^A$                  | Power set.                                                                                         |
| $A \times B$                           | Cartesian product.                                                                                 |
| $A_1 \times \dots \times A_n, A^n$     | Finite Cartesian product; $A^n$ is the $n$ -fold Cartesian power of $A$ .                          |
| $\prod_{i \in I} A_i$                  | Cartesian product of an indexed family of sets.                                                    |
| $A \sqcup B, A \dot{\cup} B$           | Disjoint union.                                                                                    |
| $\bigsqcup_{i \in I} A_i$              | Disjoint union of an indexed family of sets.                                                       |
| $A \cap B = \emptyset$                 | $A$ and $B$ are disjoint.                                                                          |
| $\{x \in A \mid P(x)\}$                | Elements $x \in A$ satisfying $P(x)$ .                                                             |
| $R \subseteq X \times Y$               | Binary relation between $X$ and $Y$ .                                                              |
| $xRy$                                  | $x$ is related to $y$ by the relation $R$ .                                                        |
| $R^{-1}$                               | Inverse relation of $R$ .                                                                          |
| $[x]_R, [x]$                           | Equivalence class of $x$ under $R$ ; the relation subscript is omitted when the relation is clear. |

| Notation                                                       | Meaning                                                                                                                   |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $x \sim y$                                                     | Equivalent elements under an understood equivalence relation.                                                             |
| $A/\sim$                                                       | Set of equivalence classes.                                                                                               |
| $\text{dom } f$                                                | Domain of $f$ .                                                                                                           |
| $\text{codom } f$                                              | Codomain of $f$ .                                                                                                         |
| $f(A)$                                                         | Image of $A$ under $f$ .                                                                                                  |
| $f^{-1}(B)$                                                    | Preimage of $B$ under $f$ ; this notation does not require $f$ to be invertible.                                          |
| $\text{im } f$                                                 | Image of the map $f$ .                                                                                                    |
| $f^{-1}: Y \rightarrow X$                                      | Inverse function of a bijection $f: X \rightarrow Y$ .                                                                    |
| $f: X \hookrightarrow Y$                                       | Injective map.                                                                                                            |
| $f: X \twoheadrightarrow Y$                                    | Surjective map.                                                                                                           |
| $f: X \xrightarrow{\sim} Y$                                    | Bijjective map.                                                                                                           |
| $f: X \xrightarrow{\sim} Y$                                    | Isomorphism; for sets, equivalently a bijection.                                                                          |
| $\mathbb{1}_A(x)$                                              | Indicator function of $A$ .                                                                                               |
| $\forall x \in A$                                              | For every $x \in A$ .                                                                                                     |
| $\exists x \in A$                                              | There exists $x \in A$ .                                                                                                  |
| $\exists! x \in A$                                             | There exists a unique $x \in A$ .                                                                                         |
| $\nexists x \in A$                                             | There does not exist $x \in A$ .                                                                                          |
| $P \Rightarrow Q$                                              | $P$ implies $Q$ in ordinary mathematical writing.                                                                         |
| $P \Leftarrow Q$                                               | $Q$ implies $P$ in ordinary mathematical writing.                                                                         |
| $P \nRightarrow Q$                                             | $P$ does not imply $Q$ .                                                                                                  |
| $P \Leftrightarrow Q$                                          | $P$ if and only if $Q$ in ordinary mathematical writing.                                                                  |
| $P \wedge Q$                                                   | Logical conjunction.                                                                                                      |
| $P \vee Q$                                                     | Logical disjunction.                                                                                                      |
| $\neg P$                                                       | Logical negation.                                                                                                         |
| $\top, \perp$                                                  | Truth and falsehood constants, respectively, in formal logic.                                                             |
| $\varphi \rightarrow \psi, \varphi \leftrightarrow \psi$       | Conditional and biconditional connectives in formal logic.                                                                |
| $\mathcal{M} \models \varphi, \mathcal{M} \not\models \varphi$ | $\mathcal{M}$ satisfies, or does not satisfy, the formula $\varphi$ .                                                     |
| $\Gamma \models \varphi, \Gamma \not\models \varphi$           | $\varphi$ is, or is not, a semantic consequence of $\Gamma$ .                                                             |
| $\Gamma \vdash \varphi, \Gamma \not\vdash \varphi$             | $\varphi$ is, or is not, syntactically derivable from $\Gamma$ .                                                          |
| $\bot$                                                         | Contradiction reached in a proof or argument; distinct from the formal falsehood constant $\perp$ .                       |
| $\therefore$                                                   | Therefore; indicates a conclusion from preceding statements.                                                              |
| $\because$                                                     | Because; indicates a reason or justification.                                                                             |
| $ A , \#A$                                                     | Cardinality; $ A $ is preferred, while $\#A$ is a common alternative, especially when emphasizing the number of elements. |
| $\aleph_0$                                                     | Cardinality of a countably infinite set.                                                                                  |
| $\mathfrak{c}$                                                 | Cardinality of the continuum.                                                                                             |

## 4 Calculus and Analysis

The differential  $d$  is upright;  $\varepsilon$  is the default small positive quantity in analysis. Function spaces take their domains in parentheses, for example  $C([a, b])$ ,  $\mathcal{R}([a, b])$ , and  $L^p(\Omega)$ .

| Notation                      | Meaning                       |
|-------------------------------|-------------------------------|
| $\lim_{x \rightarrow a} f(x)$ | Limit as $x$ approaches $a$ . |

| Notation                                                                              | Meaning                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\lim_{x \rightarrow a^-} f(x)$                                                       | Left-hand limit as $x$ approaches $a$ .                                                                                                                                                                                                                                                                                                                                                                    |
| $\lim_{x \rightarrow a^+} f(x)$                                                       | Right-hand limit as $x$ approaches $a$ .                                                                                                                                                                                                                                                                                                                                                                   |
| $\lim_{x \rightarrow \infty} f(x), \lim_{x \rightarrow -\infty} f(x)$                 | Limits as $x$ tends to positive or negative infinity.                                                                                                                                                                                                                                                                                                                                                      |
| $\lim_{x \rightarrow a} f(x) = \infty,$<br>$\lim_{x \rightarrow a} f(x) = -\infty$    | Infinite limits.                                                                                                                                                                                                                                                                                                                                                                                           |
| $\min A$                                                                              | Minimum of $A$ .                                                                                                                                                                                                                                                                                                                                                                                           |
| $\max A$                                                                              | Maximum of $A$ .                                                                                                                                                                                                                                                                                                                                                                                           |
| $\inf A$                                                                              | Infimum of $A$ .                                                                                                                                                                                                                                                                                                                                                                                           |
| $\sup A$                                                                              | Supremum of $A$ .                                                                                                                                                                                                                                                                                                                                                                                          |
| $\mathbb{R} = \mathbb{R} \cup \{-\infty, +\infty\}$                                   | Extended real line.                                                                                                                                                                                                                                                                                                                                                                                        |
| $f'(a)$                                                                               | First derivative of $f$ at $a$ .                                                                                                                                                                                                                                                                                                                                                                           |
| $f^{(n)}(a)$                                                                          | $n$ th derivative of $f$ at $a$ .                                                                                                                                                                                                                                                                                                                                                                          |
| $\frac{dy}{dx}$                                                                       | Derivative of $y$ with respect to $x$ .                                                                                                                                                                                                                                                                                                                                                                    |
| $\frac{d^k y}{dx^k}$                                                                  | $n$ th derivative of $y$ with respect to $x$ in Leibniz notation.                                                                                                                                                                                                                                                                                                                                          |
| $\frac{\partial f}{\partial x_i}, \partial_i f, f_{x_i}$                              | Partial derivative with respect to $x_i$ .                                                                                                                                                                                                                                                                                                                                                                 |
| $\frac{\partial^2 f}{\partial x_j \partial x_i}, \partial_{ji} f, f_{x_i x_j}$        | Successive or mixed partial derivative, first with respect to $x_i$ and then $x_j$ . When the relevant mixed partial derivatives are continuous, the order may be interchanged.                                                                                                                                                                                                                            |
| $D_{\mathbf{v}} f(\mathbf{a}), \partial_{\mathbf{v}} f(\mathbf{a})$                   | Directional derivative along the vector $\mathbf{v}$ , which need not be a unit vector: $D_{\mathbf{v}} f(\mathbf{a}) = \left. \frac{d}{dt} f(\mathbf{a} + t\mathbf{v}) \right _{t=0} = Df(\mathbf{a})[\mathbf{v}] = \nabla f(\mathbf{a}) \cdot \mathbf{v}$ when $f$ is differentiable. For the unit direction determined by $\mathbf{v} \neq \mathbf{0}$ , use $\mathbf{u} = \mathbf{v}/\ \mathbf{v}\ $ . |
| $df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$                              | Total differential of a scalar function of $n$ variables.                                                                                                                                                                                                                                                                                                                                                  |
| $\int f(x) dx$                                                                        | Indefinite integral; a family of antiderivatives, with an additive constant understood.                                                                                                                                                                                                                                                                                                                    |
| $\int_a^b f(x) dx$                                                                    | Definite integral over $[a, b]$ .                                                                                                                                                                                                                                                                                                                                                                          |
| $\int_a^\infty f(x) dx,$<br>$\int_{-\infty}^b f(x) dx, \int_{-\infty}^\infty f(x) dx$ | Improper integrals over unbounded intervals.                                                                                                                                                                                                                                                                                                                                                               |
| $\int_a^b f d\alpha$                                                                  | Riemann–Stieltjes integral of $f$ with respect to $\alpha$ .                                                                                                                                                                                                                                                                                                                                               |
| $P = \{x_0, \dots, x_n\},$<br>$a = x_0 < \dots < x_n = b$                             | Partition of the interval $[a, b]$ .                                                                                                                                                                                                                                                                                                                                                                       |
| $\sum_{i=1}^n f(\xi_i) \Delta x_i,$<br>$\Delta x_i = x_i - x_{i-1}$                   | Tagged Riemann sum, with $\xi_i \in [x_{i-1}, x_i]$ .                                                                                                                                                                                                                                                                                                                                                      |
| $\iint_D f dA$                                                                        | Double integral over $D$ .                                                                                                                                                                                                                                                                                                                                                                                 |
| $\iiint_\Omega f dV$                                                                  | Triple integral over $\Omega$ .                                                                                                                                                                                                                                                                                                                                                                            |
| $\int_C f(z) dz$                                                                      | Contour integral along an oriented contour $C$ .                                                                                                                                                                                                                                                                                                                                                           |
| $\oint_C f(z) dz$                                                                     | Contour integral over a closed contour $C$ .                                                                                                                                                                                                                                                                                                                                                               |

| Notation                                     | Meaning                                                                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| $-C$                                         | The contour $C$ with the opposite orientation.                                                                                           |
| $\gamma^{-1}$                                | Reversed parametrized path $\gamma$ , when unambiguous; not a functional inverse.                                                        |
| PV, P.V.                                     | Principal value; in particular, the Cauchy principal value of a singular or improper integral.                                           |
| $\sum_{k=1}^n a_k$                           | Finite sum.                                                                                                                              |
| $\sum_{k=1}^{\infty} a_k$                    | Infinite series.                                                                                                                         |
| $\prod_{k=1}^n a_k$                          | Finite product.                                                                                                                          |
| $\sum_{n=0}^{\infty} a_n(x-a)^n$             | Power series centered at $a$ .                                                                                                           |
| $\sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$ | $n$ th Taylor polynomial of $f$ about $a$ ; no single letter for the polynomial is imposed globally.                                     |
| $e^x$ , $\exp x$                             | Exponential function.                                                                                                                    |
| $\ln x$ , $\log_e x$                         | Natural logarithm of a positive real number; $\ln x$ is preferred.                                                                       |
| $\lg x$ , $\log_{10} x$                      | Logarithm to base 10.                                                                                                                    |
| $\text{lb } x$ , $\log_2 x$                  | Logarithm to base 2.                                                                                                                     |
| $\log_a x$                                   | Logarithm to base $a$ .                                                                                                                  |
| $\sin x$                                     | Sine.                                                                                                                                    |
| $\cos x$                                     | Cosine.                                                                                                                                  |
| $\tan x$                                     | Tangent.                                                                                                                                 |
| $\csc x$                                     | Cosecant.                                                                                                                                |
| $\sec x$                                     | Secant.                                                                                                                                  |
| $\cot x$                                     | Cotangent.                                                                                                                               |
| $\arcsin x$ , $\sin^{-1} x$                  | Principal inverse sine, $[-1, 1] \rightarrow [-\pi/2, \pi/2]$ .                                                                          |
| $\arccos x$ , $\cos^{-1} x$                  | Principal inverse cosine, $[-1, 1] \rightarrow [0, \pi]$ .                                                                               |
| $\arctan x$ , $\tan^{-1} x$                  | Principal inverse tangent, $\mathbb{R} \rightarrow (-\pi/2, \pi/2)$ .                                                                    |
| $\text{arccsc } x$ , $\csc^{-1} x$           | Principal inverse cosecant, defined by $\text{arccsc } x = \arcsin(1/x)$ for $ x  \geq 1$ ; its range is $[-\pi/2, 0) \cup (0, \pi/2]$ . |
| $\text{arcsec } x$ , $\sec^{-1} x$           | Principal inverse secant, defined by $\text{arcsec } x = \arccos(1/x)$ for $ x  \geq 1$ ; its range is $[0, \pi/2) \cup (\pi/2, \pi]$ .  |
| $\text{arccot } x$ , $\cot^{-1} x$           | Principal real inverse cotangent, $\mathbb{R} \rightarrow (0, \pi)$ .                                                                    |
| $\sinh x$                                    | Hyperbolic sine.                                                                                                                         |
| $\cosh x$                                    | Hyperbolic cosine.                                                                                                                       |
| $\tanh x$                                    | Hyperbolic tangent.                                                                                                                      |
| $\text{csch } x$                             | Hyperbolic cosecant.                                                                                                                     |
| $\text{sech } x$                             | Hyperbolic secant.                                                                                                                       |
| $\coth x$                                    | Hyperbolic cotangent.                                                                                                                    |

| Notation                                                                                                                                                          | Meaning                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\operatorname{arcsinh} x, \sinh^{-1} x$                                                                                                                          | Principal real inverse hyperbolic sine, $\mathbb{R} \rightarrow \mathbb{R}$ .                                                                                                                                                                                                                                                                            |
| $\operatorname{arccosh} x, \cosh^{-1} x$                                                                                                                          | Principal real inverse hyperbolic cosine, $[1, \infty) \rightarrow [0, \infty)$ .                                                                                                                                                                                                                                                                        |
| $\operatorname{arctanh} x, \tanh^{-1} x$                                                                                                                          | Principal real inverse hyperbolic tangent, $(-1, 1) \rightarrow \mathbb{R}$ .                                                                                                                                                                                                                                                                            |
| $\operatorname{arccsch} x, \operatorname{csch}^{-1} x$                                                                                                            | Principal real inverse hyperbolic cosecant, defined by $\operatorname{arccsch} x = \operatorname{arcsinh}(1/x)$ on $\mathbb{R} \setminus \{0\}$ ; its range is $\mathbb{R} \setminus \{0\}$ .                                                                                                                                                            |
| $\operatorname{arcsech} x, \operatorname{sech}^{-1} x$                                                                                                            | Principal real inverse hyperbolic secant, defined by $\operatorname{arcsech} x = \operatorname{arccosh}(1/x)$ on $(0, 1]$ ; its range is $[0, \infty)$ .                                                                                                                                                                                                 |
| $\operatorname{arccoth} x, \operatorname{coth}^{-1} x$                                                                                                            | Principal real inverse hyperbolic cotangent, defined by $\operatorname{arccoth} x = \operatorname{arctanh}(1/x)$ for $ x  > 1$ ; its range is $\mathbb{R} \setminus \{0\}$ .                                                                                                                                                                             |
| $\Gamma(z)$                                                                                                                                                       | Gamma function.                                                                                                                                                                                                                                                                                                                                          |
| $B(x, y)$                                                                                                                                                         | Beta function.                                                                                                                                                                                                                                                                                                                                           |
| $\zeta(s)$                                                                                                                                                        | Riemann zeta function.                                                                                                                                                                                                                                                                                                                                   |
| $\mathbf{J}_{\mathbf{F}}(\mathbf{a}), D\mathbf{F}(\mathbf{a})$                                                                                                    | Jacobian matrix of $\mathbf{F} = (F_1, \dots, F_m): \mathbb{R}^n \rightarrow \mathbb{R}^m$ at $\mathbf{a}$ , with $\mathbf{J}_{\mathbf{F}}(\mathbf{a}) = \left( \frac{\partial F_i}{\partial x_j}(\mathbf{a}) \right)_{i,j}$ . The notation $D\mathbf{F}(\mathbf{a})$ denotes the total derivative, identified with this matrix in standard coordinates. |
| $\det \mathbf{J}_{\mathbf{F}}(\mathbf{a}), \det D\mathbf{F}(\mathbf{a}), \left. \frac{\partial(F_1, \dots, F_n)}{\partial(x_1, \dots, x_n)} \right _{\mathbf{a}}$ | Jacobian determinant of $\mathbf{F}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ at $\mathbf{a}$ , in preferred order.                                                                                                                                                                                                                                        |
| $\operatorname{Hess} f$                                                                                                                                           | Hessian matrix of $f$ .                                                                                                                                                                                                                                                                                                                                  |
| $\nabla f, \operatorname{grad} f$                                                                                                                                 | Gradient of $f$ .                                                                                                                                                                                                                                                                                                                                        |
| $\nabla \cdot \mathbf{F}, \operatorname{div} \mathbf{F}$                                                                                                          | Divergence of $\mathbf{F}$ .                                                                                                                                                                                                                                                                                                                             |
| $\nabla \times \mathbf{F}, \operatorname{curl} \mathbf{F}$                                                                                                        | Curl of $\mathbf{F}$ .                                                                                                                                                                                                                                                                                                                                   |
| $\Delta f$                                                                                                                                                        | Laplacian of $f$ .                                                                                                                                                                                                                                                                                                                                       |
| $\mathbf{r}(t)$                                                                                                                                                   | Parametrized curve.                                                                                                                                                                                                                                                                                                                                      |
| $ds$                                                                                                                                                              | Element of arc length.                                                                                                                                                                                                                                                                                                                                   |
| $\int_C f \, ds$                                                                                                                                                  | Scalar line integral along $C$ with respect to arc length.                                                                                                                                                                                                                                                                                               |
| $\int_C \mathbf{F} \cdot d\mathbf{r}$                                                                                                                             | Line integral of the vector field $\mathbf{F}$ along $C$ .                                                                                                                                                                                                                                                                                               |
| $\iint_S f \, dS$                                                                                                                                                 | Scalar surface integral over $S$ .                                                                                                                                                                                                                                                                                                                       |
| $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$                                                                                                                       | Flux integral through $S$ .                                                                                                                                                                                                                                                                                                                              |
| $x_n \rightarrow x, \lim_{n \rightarrow \infty} x_n = x$                                                                                                          | Convergence of a sequence.                                                                                                                                                                                                                                                                                                                               |
| $(x_{n_k})$                                                                                                                                                       | Subsequence of the sequence $(x_n)$ .                                                                                                                                                                                                                                                                                                                    |
| $a_n \nearrow L$                                                                                                                                                  | Monotone increasing convergence to $L$ .                                                                                                                                                                                                                                                                                                                 |
| $a_n \searrow L$                                                                                                                                                  | Monotone decreasing convergence to $L$ .                                                                                                                                                                                                                                                                                                                 |
| $\limsup_{n \rightarrow \infty} a_n$                                                                                                                              | Limit superior of the sequence $(a_n)$ .                                                                                                                                                                                                                                                                                                                 |
| $\liminf_{n \rightarrow \infty} a_n$                                                                                                                              | Limit inferior of the sequence $(a_n)$ .                                                                                                                                                                                                                                                                                                                 |
| $f(x) \sim g(x)$                                                                                                                                                  | Asymptotic equivalence in the stated limiting regime, typically meaning $f(x)/g(x) \rightarrow 1$ .                                                                                                                                                                                                                                                      |
| $f_n \rightarrow f, f_n \xrightarrow{\text{ptw}} f$                                                                                                               | Pointwise convergence of functions on a common domain by default when no ambient topology, metric, or norm is specified. If such a structure is specified, the ordinary arrow $\rightarrow$ refers to convergence in that structure.                                                                                                                     |
| $f_n \rightrightarrows f, f_n \xrightarrow{\text{unif}} f$                                                                                                        | Uniform convergence.                                                                                                                                                                                                                                                                                                                                     |
| $f_n \xrightarrow{\text{a.e.}} f$                                                                                                                                 | Convergence almost everywhere.                                                                                                                                                                                                                                                                                                                           |

| Notation                                                  | Meaning                                                                                                                                                                                                                                |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(X, \mathcal{A}, \mu)$                                   | Measure space, where $\mathcal{A}$ is a $\sigma$ -algebra on $X$ and $\mu$ is a measure on $\mathcal{A}$ .                                                                                                                             |
| $\mathcal{A}$                                             | $\sigma$ -algebra of measurable subsets of $X$ , when the underlying set is understood.                                                                                                                                                |
| $\mathcal{B}(X)$                                          | Borel $\sigma$ -algebra of a topological space $X$ .                                                                                                                                                                                   |
| $\mu(E)$                                                  | Measure of a measurable set $E \in \mathcal{A}$ .                                                                                                                                                                                      |
| $\int_E f \, d\mu$                                        | Integral of $f$ over $E$ with respect to the measure $\mu$ .                                                                                                                                                                           |
| $f = g$ a.e., $f = g$ $\mu$ -a.e.                         | Equality almost everywhere, with the measure omitted when understood.                                                                                                                                                                  |
| $\operatorname{ess\,sup}_{x \in E} f(x)$                  | Essential supremum of $f$ on $E$ .                                                                                                                                                                                                     |
| $\operatorname{ess\,inf}_{x \in E} f(x)$                  | Essential infimum of $f$ on $E$ .                                                                                                                                                                                                      |
| $C(\Omega)$ , $C^0(\Omega)$                               | Continuous functions on $\Omega$ .                                                                                                                                                                                                     |
| $C^k(\Omega)$                                             | $k$ times continuously differentiable functions on $\Omega$ .                                                                                                                                                                          |
| $C^\infty(\Omega)$                                        | Smooth functions on $\Omega$ .                                                                                                                                                                                                         |
| $\operatorname{supp} f$                                   | Support of $f$ , the closure of the set on which $f$ is nonzero.                                                                                                                                                                       |
| $C_c(\Omega)$ , $C_c^\infty(\Omega)$                      | Continuous and smooth functions on $\Omega$ with compact support, respectively.                                                                                                                                                        |
| $\mathcal{R}([a, b])$                                     | Riemann-integrable functions on $[a, b]$ .                                                                                                                                                                                             |
| $L^p(\Omega)$ , $L^p(X, \mu)$                             | Lebesgue $L^p$ spaces; the measure is displayed when it is not understood.                                                                                                                                                             |
| $L^\infty(\Omega)$ , $L^\infty(X, \mu)$                   | Lebesgue spaces of essentially bounded functions.                                                                                                                                                                                      |
| $\ell^p$                                                  | Space of scalar sequences $(x_n)$ with $\sum_{n=1}^\infty  x_n ^p < \infty$ , for $1 \leq p < \infty$ .                                                                                                                                |
| $\ell^\infty$                                             | Space of bounded scalar sequences.                                                                                                                                                                                                     |
| $\ f\ _\infty$                                            | Supremum norm $\sup_{\mathbf{x} \in \Omega}  f(\mathbf{x}) $ for an actual bounded function on a Euclidean domain, especially in bounded- or continuous-function spaces.                                                               |
| $\ f\ _{L^p(X, \mu)}$                                     | $L^p$ -norm $\left( \int_X  f ^p \, d\mu \right)^{1/p}$ for $1 \leq p < \infty$ on a measure space $(X, \mathcal{A}, \mu)$ .                                                                                                           |
| $\ f\ _{L^p(\Omega)}$ , $\ f\ _p$                         | $L^p$ -norm $\left( \int_\Omega  f(\mathbf{x}) ^p \, d\mathbf{x} \right)^{1/p}$ for $1 \leq p < \infty$ on a Euclidean domain when the underlying measure is understood.                                                               |
| $\ f\ _{L^\infty(\Omega)}$                                | $L^\infty$ -norm $\operatorname{ess\,sup}_{\mathbf{x} \in \Omega}  f(\mathbf{x}) $ on a Euclidean domain; this notation is preferred when distinguishing the essential-supremum norm from an ordinary supremum norm.                   |
| $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}_0^n$ | Multi-index; $\alpha$ is not treated as a vector for typographic purposes.                                                                                                                                                             |
| $ \alpha  = \alpha_1 + \dots + \alpha_n$                  | Order and factorial of a multi-index.                                                                                                                                                                                                  |
| $\alpha! = \alpha_1! \dots \alpha_n!$                     |                                                                                                                                                                                                                                        |
| $\mathbf{x}^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$ | Multi-index monomial for $\mathbf{x} = (x_1, \dots, x_n)$ .                                                                                                                                                                            |
| $D^\alpha f$ , $\partial^\alpha f$                        | Mixed partial derivative specified by the multi-index $\alpha$ .                                                                                                                                                                       |
| $B_r(x)$                                                  | Open ball of radius $r$ centered at $x$ .                                                                                                                                                                                              |
| $\overline{B}_r(x) = \{y \mid d(x, y) \leq r\}$           | Closed ball of radius $r$ centered at $x$ in a metric space $(X, d)$ . The bar is part of the closed-ball notation; in an arbitrary metric space, $\overline{B}_r(x)$ need not equal the closure $\overline{B_r(x)}$ of the open ball. |
| $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$       | Extended complex plane, equivalently the Riemann sphere.                                                                                                                                                                               |
| $i^2 = -1$                                                | Imaginary unit.                                                                                                                                                                                                                        |

| Notation                                                            | Meaning                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $z = x + iy$                                                        | Complex number with real part $x$ and imaginary part $y$ .                                                                                                                                                                                                                |
| $\operatorname{Re} z, \Re z$                                        | Real part of $z$ .                                                                                                                                                                                                                                                        |
| $\operatorname{Im} z, \Im z$                                        | Imaginary part of $z$ ; distinct from lowercase $\operatorname{im}$ for image.                                                                                                                                                                                            |
| $\bar{z}$                                                           | Complex conjugate.                                                                                                                                                                                                                                                        |
| $ z $                                                               | Complex modulus.                                                                                                                                                                                                                                                          |
| $\arg z$                                                            | Argument of a nonzero complex number $z$ ; its values differ by integer multiples of $2\pi$ .                                                                                                                                                                             |
| $\operatorname{Arg} z$                                              | Principal argument of $z \neq 0$ , with $-\pi < \operatorname{Arg} z \leq \pi$ .                                                                                                                                                                                          |
| $\log z, \ln z$                                                     | General multivalued complex logarithm for $z \neq 0$ ; $\log z$ is preferred and $\log z = \{\ln  z  + i(\operatorname{Arg} z + 2\pi k) \mid k \in \mathbb{Z}\}$ .                                                                                                        |
| $\operatorname{Log} z, \operatorname{Ln} z$                         | Principal complex logarithm for $z \neq 0$ ; $\operatorname{Log} z$ is preferred and $\operatorname{Log} z = \ln  z  + i \operatorname{Arg} z$ . On $\mathbb{C} \setminus (-\infty, 0]$ , it is the analytic principal branch, with $-\pi < \operatorname{Arg} z < \pi$ . |
| $\operatorname{Res}(f; a), \operatorname{Res}_{z=a} f(z)$           | Residue of $f$ at a finite point $a$ .                                                                                                                                                                                                                                    |
| $\operatorname{Res}(f; \infty), \operatorname{Res}_{z=\infty} f(z)$ | Residue of $f$ at infinity.                                                                                                                                                                                                                                               |

## 5 Linear Algebra

| Notation                                                         | Meaning                                                                                                                                       |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| $M_{m \times n}(F), F^{m \times n}$                              | Space of $m \times n$ matrices over the field $F$ ; $M_{m \times n}(F)$ is preferred.                                                         |
| $M_n(F)$                                                         | Space (and matrix algebra) of $n \times n$ matrices over $F$ .                                                                                |
| $\mathbf{A} = (a_{ij}), \mathbf{A} = [a_{ij}]$                   | Matrix with $(i, j)$ -entry $a_{ij}$ ; parentheses are preferred, with square brackets a standard alternative.                                |
| $\mathbf{A}^\top$                                                | Transpose of $\mathbf{A}$ .                                                                                                                   |
| $\mathbf{A}^H$                                                   | Conjugate transpose of $\mathbf{A}$ .                                                                                                         |
| $\mathbf{A}^{-1}$                                                | Inverse of $\mathbf{A}$ .                                                                                                                     |
| $\mathbf{I}_n, \mathbf{I}$                                       | Identity matrix; the dimension subscript is preferred when readily available and useful.                                                      |
| $\mathbf{0}$                                                     | Zero vector or zero matrix, with dimension understood.                                                                                        |
| $\mathbf{e}_i$                                                   | $i$ th standard basis vector.                                                                                                                 |
| $\mathbf{1}$                                                     | All-ones vector.                                                                                                                              |
| $\mathbf{D}$                                                     | Diagonal matrix when a local name is required.                                                                                                |
| $\operatorname{diag}(d_1, \dots, d_n)$                           | Diagonal matrix with entries $d_1, \dots, d_n$ .                                                                                              |
| $\mathbf{P}$                                                     | Projection matrix when a local name is required.                                                                                              |
| $\det(\mathbf{A}),  \mathbf{A} $                                 | Determinant; $\det(\mathbf{A})$ is preferred.                                                                                                 |
| $\operatorname{rk}(\mathbf{A}), \operatorname{rank}(\mathbf{A})$ | Rank; $\operatorname{rk}(\mathbf{A})$ is preferred.                                                                                           |
| $\operatorname{tr}(\mathbf{A})$                                  | Trace.                                                                                                                                        |
| $\operatorname{adj}(\mathbf{A})$                                 | Adjugate matrix.                                                                                                                              |
| $M_{ij}$                                                         | Minor of the $(i, j)$ -entry, when the matrix is understood.                                                                                  |
| $C_{ij} = (-1)^{i+j} M_{ij}$                                     | Cofactor of the $(i, j)$ -entry.                                                                                                              |
| $\operatorname{Col}(\mathbf{A})$                                 | Column space.                                                                                                                                 |
| $\operatorname{Row}(\mathbf{A})$                                 | Row space.                                                                                                                                    |
| $\ker \mathbf{A}, \operatorname{Null}(\mathbf{A})$               | Null space of $\mathbf{A}$ ; $\ker \mathbf{A}$ is preferred, while $\operatorname{Null}(\mathbf{A})$ is a common matrix-oriented alternative. |
| $V, W$                                                           | Vector spaces, usually over an understood field $F$ .                                                                                         |
| $\dim V$                                                         | Dimension of $V$ .                                                                                                                            |
| $\operatorname{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$         | Span of the listed vectors.                                                                                                                   |

| Notation                                                                          | Meaning                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $U \oplus V$                                                                      | Direct sum.                                                                                                                                                                                                              |
| $\ker T$                                                                          | Kernel of a linear map $T$ .                                                                                                                                                                                             |
| $\operatorname{im} T$                                                             | Image of a linear map $T$ .                                                                                                                                                                                              |
| $\mathcal{B} = (\mathbf{b}_1, \dots, \mathbf{b}_n)$                               | Ordered basis.                                                                                                                                                                                                           |
| $[\mathbf{v}]_{\mathcal{B}}$                                                      | Coordinate vector of $\mathbf{v}$ with respect to the ordered basis $\mathcal{B}$ .                                                                                                                                      |
| $[T]_{\mathcal{C} \leftarrow \mathcal{B}}$                                        | Matrix of $T: V \rightarrow W$ with respect to domain basis $\mathcal{B}$ and codomain basis $\mathcal{C}$ ; thus $[T(\mathbf{v})]_{\mathcal{C}} = [T]_{\mathcal{C} \leftarrow \mathcal{B}}[\mathbf{v}]_{\mathcal{B}}$ . |
| $[T]_{\mathcal{B}}$                                                               | Matrix of an endomorphism $T: V \rightarrow V$ when the same basis $\mathcal{B}$ is used in the domain and codomain.                                                                                                     |
| $\mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}$                                 | Change-of-coordinate matrix from $\mathcal{B}$ -coordinates to $\mathcal{C}$ -coordinates; $[\mathbf{v}]_{\mathcal{C}} = \mathbf{P}_{\mathcal{C} \leftarrow \mathcal{B}}[\mathbf{v}]_{\mathcal{B}}$ .                    |
| $V^*$                                                                             | Dual space of $V$ .                                                                                                                                                                                                      |
| $\langle \mathbf{u}, \mathbf{v} \rangle, (\mathbf{u}, \mathbf{v})$                | Inner product; angle brackets are preferred. Over complex spaces, the inner product is conjugate-linear in the first argument and linear in the second.                                                                  |
| $\mathbf{u} \cdot \mathbf{v}, \mathbf{u}^T \mathbf{v}$                            | Standard Euclidean inner product on $\mathbb{R}^n$ .                                                                                                                                                                     |
| $\mathbf{u}^H \mathbf{v}$                                                         | Standard Hermitian inner product on $\mathbb{C}^n$ .                                                                                                                                                                     |
| $\ \mathbf{v}\ $                                                                  | Norm of $\mathbf{v}$ .                                                                                                                                                                                                   |
| $\mathbf{u} \perp \mathbf{v}$                                                     | Orthogonality.                                                                                                                                                                                                           |
| $U^\perp$                                                                         | Orthogonal complement of the subspace $U$ .                                                                                                                                                                              |
| $\operatorname{proj}_U \mathbf{x}$                                                | Orthogonal projection of $\mathbf{x}$ onto the subspace $U$ .                                                                                                                                                            |
| $\operatorname{proj}_{\mathbf{v}} \mathbf{x}$                                     | Orthogonal projection of $\mathbf{x}$ onto $\operatorname{span}(\mathbf{v})$ .                                                                                                                                           |
| $\mathbf{u} \times \mathbf{v}$                                                    | Cross product in $\mathbb{R}^3$ .                                                                                                                                                                                        |
| $\mathbf{A}^T = \mathbf{A}$                                                       | Real symmetric matrix.                                                                                                                                                                                                   |
| $\mathbf{A}^T = -\mathbf{A}$                                                      | Real skew-symmetric matrix.                                                                                                                                                                                              |
| $\mathbf{A}^H = \mathbf{A}$                                                       | Hermitian matrix.                                                                                                                                                                                                        |
| $\mathbf{A}^H = -\mathbf{A}$                                                      | Skew-Hermitian matrix.                                                                                                                                                                                                   |
| $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$                                            | Real orthogonal matrix $\mathbf{Q}$ .                                                                                                                                                                                    |
| $\mathbf{U}^H \mathbf{U} = \mathbf{I}$                                            | Unitary matrix $\mathbf{U}$ .                                                                                                                                                                                            |
| $\mathbf{A} \mathbf{A}^H = \mathbf{A}^H \mathbf{A}$                               | Normal matrix.                                                                                                                                                                                                           |
| $\mathbf{P}^2 = \mathbf{P}$                                                       | Projection matrix.                                                                                                                                                                                                       |
| $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^T$                                        | Orthogonal projection matrix over $\mathbb{R}$ .                                                                                                                                                                         |
| $\mathbf{P}^2 = \mathbf{P} = \mathbf{P}^H$                                        | Orthogonal projection matrix over $\mathbb{C}$ .                                                                                                                                                                         |
| $\chi_{\mathbf{A}}(t), p_{\mathbf{A}}(t)$                                         | Characteristic polynomial of $\mathbf{A}$ , with $\chi_{\mathbf{A}}(t) = \det(t\mathbf{I} - \mathbf{A})$ ; $\chi_{\mathbf{A}}$ is preferred.                                                                             |
| $m_{\mathbf{A}}(t), \mu_{\mathbf{A}}(t)$                                          | Minimal polynomial of $\mathbf{A}$ ; $m_{\mathbf{A}}$ is preferred.                                                                                                                                                      |
| $\lambda, \lambda_i$                                                              | Eigenvalue of $\mathbf{A}$ .                                                                                                                                                                                             |
| $\mathbf{v}, \mathbf{v}_i$                                                        | Eigenvector of $\mathbf{A}$ .                                                                                                                                                                                            |
| $E_{\lambda}(\mathbf{A}), V_{\lambda}(\mathbf{A})$                                | Eigenspace associated with $\lambda$ ; $E_{\lambda}(\mathbf{A}) = \ker(\mathbf{A} - \lambda\mathbf{I})$ .                                                                                                                |
| $\sigma(\mathbf{A}), \operatorname{spec}(\mathbf{A})$                             | Spectrum of $\mathbf{A}$ ; $\sigma(\mathbf{A})$ is preferred.                                                                                                                                                            |
| $\rho(\mathbf{A})$                                                                | Spectral radius.                                                                                                                                                                                                         |
| $\mathbf{P}_{\lambda}$                                                            | Spectral projection associated with the eigenvalue $\lambda$ , when defined.                                                                                                                                             |
| $\mathbf{A} = \mathbf{P} \mathbf{D} \mathbf{P}^{-1}$                              | Diagonalization of a diagonalizable matrix.                                                                                                                                                                              |
| $\mathbf{A} = \mathbf{Q} \mathbf{D} \mathbf{Q}^T$                                 | Orthogonal diagonalization of a real symmetric matrix, with $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$ .                                                                                                                     |
| $\mathbf{A} = \mathbf{U} \mathbf{D} \mathbf{U}^H$                                 | Unitary diagonalization of a complex normal matrix, with $\mathbf{U}^H \mathbf{U} = \mathbf{I}$ .                                                                                                                        |
| $\mathbf{A} = \sum_{\lambda \in \sigma(\mathbf{A})} \lambda \mathbf{P}_{\lambda}$ | Spectral decomposition of a diagonalizable matrix, with the sum over distinct eigenvalues.                                                                                                                               |

| Notation                                                        | Meaning                                                                                                                                   |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| $e^{\mathbf{A}}, \exp(\mathbf{A})$                              | Matrix exponential; $e^{t\mathbf{A}}$ is the standard time-dependent form.                                                                |
| $\mathbf{A} = \mathbf{L}\mathbf{U}$                             | LU factorization, with $\mathbf{L}$ lower triangular and $\mathbf{U}$ upper triangular.                                                   |
| $\mathbf{A} = \mathbf{Q}\mathbf{R}$                             | QR factorization, with $\mathbf{Q}$ orthogonal or unitary and $\mathbf{R}$ upper triangular.                                              |
| $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathbf{H}}$ | Singular value decomposition (SVD); over $\mathbb{R}$ , $\mathbf{V}^{\mathbf{H}} = \mathbf{V}^{\mathbf{T}}$ .                             |
| $\sigma_i(\mathbf{A})$                                          | $i$ th singular value of $\mathbf{A}$ .                                                                                                   |
| $\mathbf{A}^{\dagger}$                                          | Moore–Penrose pseudoinverse of $\mathbf{A}$ .                                                                                             |
| $\mathbf{A} \succ 0$                                            | Positive definite Hermitian matrix (real symmetric in the real case).                                                                     |
| $\mathbf{A} \succeq 0$                                          | Positive semidefinite Hermitian matrix.                                                                                                   |
| $\mathbf{A} \prec 0$                                            | Negative definite Hermitian matrix.                                                                                                       |
| $\mathbf{A} \preceq 0$                                          | Negative semidefinite Hermitian matrix.                                                                                                   |
| $\mathbf{A} \succeq \mathbf{B}$                                 | Löwner order: $\mathbf{A} - \mathbf{B} \succeq 0$ , for Hermitian matrices of the same size.                                              |
| $\mathbf{A} = \mathbf{L}\mathbf{L}^{\mathbf{H}}$                | Cholesky factorization of a positive definite Hermitian matrix; over $\mathbb{R}$ , $\mathbf{L}^{\mathbf{H}} = \mathbf{L}^{\mathbf{T}}$ . |

## 6 Differential Equations

| Notation                                            | Meaning                                                                                                                                 |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| $y'$                                                | First derivative with respect to the understood independent variable.                                                                   |
| $y''$                                               | Second derivative.                                                                                                                      |
| $\dot{x}$                                           | First derivative of $x$ with respect to time.                                                                                           |
| $\ddot{x}$                                          | Second derivative of $x$ with respect to time.                                                                                          |
| $F(t, y, y', \dots, y^{(n)}) = 0$                   | General $n$ th-order ordinary differential equation.                                                                                    |
| $y^{(n)} = f(t, y, \dots, y^{(n-1)})$               | $n$ th-order ordinary differential equation in normal form.                                                                             |
| $y(t_0) = y_0$                                      | Initial condition for a scalar differential equation.                                                                                   |
| $y' + p(t)y = g(t)$                                 | Standard form of a first-order linear differential equation.                                                                            |
| $\mu(t)$                                            | Integrating factor; for $y' + p(t)y = g(t)$ , one may take $\mu(t) = \exp\left(\int p(t) \, dt\right)$ up to a nonzero constant factor. |
| $M(x, y) \, dx + N(x, y) \, dy = 0$                 | Differential form of a first-order differential equation.                                                                               |
| $M_x, M_y, N_x, N_y$                                | Subscript notation for the corresponding partial derivatives.                                                                           |
| $L[y]$                                              | Linear differential operator applied to $y$ ; $\mathcal{L}$ is reserved here for the Laplace transform.                                 |
| $L[y] = \sum_{k=0}^n a_k(t)y^{(k)}$                 | General $n$ th-order linear differential operator, with $a_n \neq 0$ .                                                                  |
| $L[y] = 0, \, L[y] = g$                             | Homogeneous and nonhomogeneous linear differential equations, respectively.                                                             |
| $y_h, \, y_c$                                       | Homogeneous or complementary part of a solution; $y_h$ is preferred.                                                                    |
| $y_p$                                               | Particular solution.                                                                                                                    |
| $c_1, c_2, \dots$                                   | Arbitrary constants in a general solution.                                                                                              |
| $\sum_{k=0}^n a_k r^k = 0$                          | Characteristic equation for a constant-coefficient homogeneous linear differential equation.                                            |
| $r_1, \dots, r_n$                                   | Characteristic roots; multiplicities are counted when relevant.                                                                         |
| $W(y_1, \dots, y_n)(t),$<br>$W[y_1, \dots, y_n](t)$ | Wronskian of $y_1, \dots, y_n$ ; parentheses are preferred.                                                                             |

| Notation                                                       | Meaning                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $y = \sum_{n=0}^{\infty} a_n (x - x_0)^n$                      | Power-series form about the ordinary point $x_0$ .                                                                                                                                                                                                                                                                                                    |
| $y = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r}$                  | Frobenius-series form about a regular singular point $x_0$ ; $r$ denotes an indicial exponent.                                                                                                                                                                                                                                                        |
| $\mathbf{x}' = \mathbf{f}(t, \mathbf{x})$                      | General first-order system.                                                                                                                                                                                                                                                                                                                           |
| $\mathbf{x}(t_0) = \mathbf{x}_0$                               | Initial condition for a first-order system.                                                                                                                                                                                                                                                                                                           |
| $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$           | Linear first-order system in state-vector form.                                                                                                                                                                                                                                                                                                       |
| $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$      |                                                                                                                                                                                                                                                                                                                                                       |
| $\mathbf{x}' = \mathbf{f}(\mathbf{x})$                         | Autonomous first-order system.                                                                                                                                                                                                                                                                                                                        |
| $\mathbf{x}_e$                                                 | Equilibrium point; $\mathbf{f}(\mathbf{x}_e) = \mathbf{0}$ .                                                                                                                                                                                                                                                                                          |
| $\Phi(t)$                                                      | Fundamental matrix of a linear system.                                                                                                                                                                                                                                                                                                                |
| $\mathcal{L}\{f(t)\}(s)$ ,<br>$\mathcal{L}[f](s)$              | Laplace transform of $f$ ; braces are preferred.                                                                                                                                                                                                                                                                                                      |
| $\mathcal{L}^{-1}\{F(s)\}(t)$ ,<br>$\mathcal{L}^{-1}[F](t)$    | Inverse Laplace transform of $F$ ; braces are preferred.                                                                                                                                                                                                                                                                                              |
| $(f * g)(t)$                                                   | One-sided convolution, $(f * g)(t) := \int_0^t f(\tau)g(t - \tau) \, d\tau$ .                                                                                                                                                                                                                                                                         |
| $H(t - a)$ , $u(t - a)$                                        | Heaviside or unit-step function shifted to $t = a$ ; $H$ is preferred, and its value at the jump is specified when relevant.                                                                                                                                                                                                                          |
| $\delta(t)$ , $\delta(t - a)$                                  | Dirac delta distribution, or unit impulse, centered at 0 or shifted to $t = a$ .                                                                                                                                                                                                                                                                      |
| $t_n = t_0 + nh$                                               | Uniform numerical grid with step size $h > 0$ .                                                                                                                                                                                                                                                                                                       |
| $y_n \approx y(t_n)$                                           | Numerical approximation to the exact solution at $t_n$ .                                                                                                                                                                                                                                                                                              |
| $u_x, u_t, u_{xx}, u_{xy}, u_{tt}, \dots$                      | Subscript notation for partial derivatives of a scalar function $u$ .                                                                                                                                                                                                                                                                                 |
| $\partial_{\mathbf{n}}u$                                       | Normal derivative, $\partial_{\mathbf{n}}u := \nabla u \cdot \mathbf{n}$ , for a unit normal vector $\mathbf{n}$ .                                                                                                                                                                                                                                    |
| $u _{\partial\Omega} = g$                                      | Dirichlet boundary condition on $\partial\Omega$ .                                                                                                                                                                                                                                                                                                    |
| $\partial_{\mathbf{n}}u _{\partial\Omega} = g$                 | Neumann boundary condition on $\partial\Omega$ .                                                                                                                                                                                                                                                                                                      |
| $\alpha u + \beta \partial_{\mathbf{n}}u = g$                  | Robin boundary condition on $\partial\Omega$ .                                                                                                                                                                                                                                                                                                        |
| $a_n$ , $b_n$                                                  | Fourier coefficients for the real $2L$ -periodic convention                                                                                                                                                                                                                                                                                           |
|                                                                | $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$ .                                                                                                                                                                                                                                |
| $a_n$                                                          | $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} \, dx$ for $n \geq 0$ .                                                                                                                                                                                                                                                                     |
| $b_n$                                                          | $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} \, dx$ for $n \geq 1$ .                                                                                                                                                                                                                                                                     |
| $\hat{f}(\boldsymbol{\xi})$ , $\mathcal{F}f(\boldsymbol{\xi})$ | Fourier transform under the normalization<br>$\hat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} \, d\mathbf{x}$ ; inversion uses<br>$f(\mathbf{x}) = \int_{\mathbb{R}^n} \hat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} \, d\boldsymbol{\xi}$ whenever applicable. |
| $-(py')' + qy = \lambda wy$                                    | Sturm–Liouville eigenvalue equation, where $p$ , $q$ , and $w$ are functions of $x$ .                                                                                                                                                                                                                                                                 |
| $\lambda_n$ , $\phi_n$                                         | Sturm–Liouville eigenvalue and corresponding eigenfunction.                                                                                                                                                                                                                                                                                           |
| $\langle f, g \rangle_w$                                       | Weighted inner product, $\langle f, g \rangle_w := \int_a^b \overline{f(x)} g(x) w(x) \, dx$ .                                                                                                                                                                                                                                                        |
| $\langle \phi_m, \phi_n \rangle_w = \delta_{mn}$               | Weighted orthonormality of eigenfunctions.                                                                                                                                                                                                                                                                                                            |
| $G(x, \boldsymbol{\xi})$ , $G(\mathbf{x}, \boldsymbol{\xi})$   | Green's function; arguments are scalar or vector as appropriate.                                                                                                                                                                                                                                                                                      |
| $\text{Ai}(x)$ , $\text{Bi}(x)$                                | Airy functions.                                                                                                                                                                                                                                                                                                                                       |

| Notation                            | Meaning                                                                   |
|-------------------------------------|---------------------------------------------------------------------------|
| $J_\nu(x)$                          | Bessel function of the first kind of order $\nu$ .                        |
| $Y_\nu(x)$ , $N_\nu(x)$             | Bessel function of the second kind of order $\nu$ ; $Y_\nu$ is preferred. |
| $I_\nu(x)$ , $K_\nu(x)$             | Modified Bessel functions of the first and second kinds.                  |
| $H_\nu^{(1)}(x)$ , $H_\nu^{(2)}(x)$ | Hankel functions of the first and second kinds.                           |
| $P_n(x)$                            | Legendre polynomial of degree $n$ .                                       |
| $P_\nu(x)$ , $Q_\nu(x)$             | Legendre functions of the first and second kinds.                         |
| $P_\ell^m(x)$                       | Associated Legendre function for integer degree $\ell$ and order $m$ .    |
| $H_n(x)$ , $\text{He}_n(x)$         | Physicists' and probabilists' Hermite polynomials, respectively.          |
| $L_n^{(\alpha)}(x)$ , $L_n(x)$      | Generalized and ordinary Laguerre polynomials; $L_n(x) = L_n^{(0)}(x)$ .  |
| $T_n(x)$ , $U_n(x)$                 | Chebyshev polynomials of the first and second kinds.                      |
| $P_n^{(\alpha, \beta)}(x)$          | Jacobi polynomial.                                                        |
| $C_n^{(\lambda)}(x)$                | Gegenbauer (ultraspherical) polynomial.                                   |
| $\text{erf}(x)$ , $\text{erfc}(x)$  | Error function and complementary error function.                          |
| ${}_2F_1(a, b; c; z)$               | Gauss hypergeometric function.                                            |

## 7 Combinatorics and Discrete Mathematics

| Notation                                                   | Meaning                                                                                                                        |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| $[n] = \{1, 2, \dots, n\}$                                 | Standard $n$ -element set, especially in combinatorics; $[0] = \emptyset$ when needed.                                         |
| $\binom{n}{r}$ , ${}_nC_r$                                 | Number of $r$ -element subsets of an $n$ -element set.                                                                         |
| $\left(\binom{n}{r}\right)$ , ${}_nH_r$                    | $r$ -combinations with repetition from $n$ types; equals $\binom{n+r-1}{r}$ .                                                  |
| $\binom{n}{n_1, \dots, n_k} = \frac{n!}{n_1! \cdots n_k!}$ | Multinomial coefficient, for nonnegative integers $n_1, \dots, n_k$ with $n_1 + \cdots + n_k = n$ .                            |
| $n!$                                                       | Factorial.                                                                                                                     |
| $n^{\underline{k}}$                                        | Falling factorial, $n(n-1) \cdots (n-k+1)$ .                                                                                   |
| $n^{\overline{k}}$                                         | Rising factorial, $n(n+1) \cdots (n+k-1)$ .                                                                                    |
| $(a)_k$                                                    | Rising Pochhammer symbol $a(a+1) \cdots (a+k-1)$ in special-functions contexts; it is not used here for the falling factorial. |
| $(X, \preceq)$                                             | Partially ordered set (poset).                                                                                                 |
| $\prec, \succeq, \succ$                                    | Strict and reverse order relations associated with $\preceq$ .                                                                 |
| $\leq_{\text{lex}}$                                        | Lexicographic order.                                                                                                           |
| $x \lessdot y$                                             | Cover relation: $y$ covers $x$ .                                                                                               |
| $B_n$                                                      | Boolean lattice (Boolean poset) on an $n$ -element ground set.                                                                 |
| $G = (V, E)$                                               | Graph with vertex set $V$ and edge set $E$ .                                                                                   |
| $V(G)$                                                     | Vertex set of $G$ .                                                                                                            |
| $E(G)$                                                     | Edge set of $G$ .                                                                                                              |
| $N(v)$ , $\Gamma(v)$                                       | Neighborhood of vertex $v$ .                                                                                                   |
| $\deg_G(v)$ , $\deg(v)$                                    | Degree of vertex $v$ ; the graph subscript may be omitted when clear.                                                          |
| $\deg^+(v)$ , $\deg^-(v)$                                  | Out-degree and in-degree of $v$ in a directed graph.                                                                           |
| $d_G(u, v)$ , $d(u, v)$                                    | Graph distance between vertices $u$ and $v$ .                                                                                  |
| $G[S]$                                                     | Subgraph of $G$ induced by the vertex set $S \subseteq V(G)$ .                                                                 |
| $G - v$ , $G - S$                                          | Graph obtained by deleting the vertex $v$ , or all vertices in $S$ , together with their incident edges.                       |
| $G - e$                                                    | Graph obtained by deleting the edge $e$ .                                                                                      |

| Notation                                         | Meaning                                                                                                               |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| $G/e$                                            | Graph obtained by contracting the edge $e$ .                                                                          |
| $\overline{G}$                                   | Complement of the graph $G$ .                                                                                         |
| $K_n$                                            | Complete graph on $n$ vertices.                                                                                       |
| $K_{m,n}$                                        | Complete bipartite graph with parts of sizes $m$ and $n$ .                                                            |
| $P_n$                                            | Path graph on $n$ vertices.                                                                                           |
| $C_n$                                            | Cycle graph on $n$ vertices.                                                                                          |
| $\mathbf{A}(G)$ , $\mathbf{A}_G$                 | Adjacency matrix of $G$ .                                                                                             |
| $\mathbf{D}(G)$ , $\mathbf{D}_G$                 | Degree matrix of $G$ , the diagonal matrix of vertex degrees in the chosen vertex order.                              |
| $\mathbf{L}(G)$ , $\mathbf{L}_G$                 | Graph Laplacian, usually $\mathbf{D}(G) - \mathbf{A}(G)$ .                                                            |
| $\chi(G)$                                        | Chromatic number of $G$ .                                                                                             |
| $\tau(G)$ , $T(G)$                               | Number of spanning trees of $G$ ; $\tau(G)$ is preferred.                                                             |
| $R(k, \ell)$ , $r(k, \ell)$ ,<br>$R(k)$ , $r(k)$ | Ramsey numbers; uppercase $R$ is preferred, and $R(k) = R(k, k)$ in the diagonal case.                                |
| $A(x) = \sum_{n=0}^{\infty} a_n x^n$             | Ordinary generating function of the sequence $(a_n)_{n \geq 0}$ .                                                     |
| $\sum_{n=0}^{\infty} a_n \frac{x^n}{n!}$         | Exponential generating function of the sequence $(a_n)_{n \geq 0}$ ; the function name is chosen locally when needed. |
| $[x^n]F(x)$                                      | Coefficient of $x^n$ in the formal power series or generating function $F(x)$ .                                       |

## 8 Probability and Statistics

Probability and expectation are written in blackboard bold by default. Probabilities use parentheses and expectations square brackets by default. Parameter symbols are local to the relevant context: for gamma distributions,  $\beta$  denotes a rate parameter and  $\theta$  a scale parameter, so  $\theta = 1/\beta$ , while in hypothesis testing  $\beta$  denotes a Type II error probability. The geometric distribution counts the number of trials up to and including the first success.

| Notation                                  | Meaning                                                                                                                           |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $(\Omega, \mathcal{F}, \mathbb{P})$       | Probability space, with sample space $\Omega$ , event $\sigma$ -algebra $\mathcal{F}$ , and probability measure $\mathbb{P}$ .    |
| $\omega \in \Omega$                       | Outcome or sample point.                                                                                                          |
| $A \in \mathcal{F}$                       | Event.                                                                                                                            |
| $X: \Omega \rightarrow \mathbb{R}$        | Real-valued random variable; measurability from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is understood. |
| $\mathbb{P}(A)$ , $P(A)$                  | Probability of event $A$ .                                                                                                        |
| $\mathbb{P}(A \mid B)$ , $P(A \mid B)$    | Conditional probability of $A$ given $B$ .                                                                                        |
| $X \perp\!\!\!\perp Y$                    | $X$ and $Y$ are independent.                                                                                                      |
| $X \perp\!\!\!\perp Y \mid Z$             | $X$ and $Y$ are conditionally independent given $Z$ .                                                                             |
| $p_X(x) = \mathbb{P}(X = x)$              | Probability mass function (pmf) of a discrete random variable $X$ .                                                               |
| $f_X(x)$                                  | Probability density function (pdf) of a continuous random variable $X$ .                                                          |
| $F_X(x) = \mathbb{P}(X \leq x)$           | Cumulative distribution function (cdf) of $X$ .                                                                                   |
| $p_{X,Y}(x, y)$ , $f_{X,Y}(x, y)$         | Joint pmf or joint pdf of $(X, Y)$ , according to context.                                                                        |
| $p_{X Y}(x \mid y)$ , $f_{X Y}(x \mid y)$ | Conditional pmf or conditional pdf of $X$ given $Y = y$ , according to context.                                                   |

| Notation                                                        | Meaning                                                                                                                                                                                    |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F^{-1}(p)$                                                     | Generalized inverse of a cdf $F$ , $F^{-1}(p) = \inf\{x \in \mathbb{R} \mid F(x) \geq p\}$ ; the $p$ -quantile, for $0 < p < 1$ .                                                          |
| $\mathbb{E}[X]$ , $E[X]$                                        | Expectation of random variable $X$ .                                                                                                                                                       |
| $\mathbb{E}[X \mid Y]$ , $E[X \mid Y]$                          | Conditional expectation of $X$ given $Y$ .                                                                                                                                                 |
| $\mathbb{E}[X \mid \mathcal{G}]$ , $E[X \mid \mathcal{G}]$      | Conditional expectation of $X$ with respect to a sub- $\sigma$ -algebra $\mathcal{G} \subseteq \mathcal{F}$ .                                                                              |
| $\text{Var}(X)$                                                 | Variance of $X$ .                                                                                                                                                                          |
| $\sigma_X = \sqrt{\text{Var}(X)}$                               | Population standard deviation of $X$ .                                                                                                                                                     |
| $\text{Cov}(X, Y)$                                              | Covariance of $X$ and $Y$ .                                                                                                                                                                |
| $\rho_{X,Y}$                                                    | Population correlation coefficient of $X$ and $Y$ .                                                                                                                                        |
| $r$                                                             | Sample correlation coefficient, when the sample is understood.                                                                                                                             |
| $\varphi_X(t) = \mathbb{E}[e^{itX}]$                            | Characteristic function of $X$ .                                                                                                                                                           |
| $M_X(t) = \mathbb{E}[e^{tX}]$                                   | Moment-generating function (mgf) of $X$ , when finite in a neighborhood of 0.                                                                                                              |
| $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$                        | Sample mean as a statistic of a random sample $X_1, \dots, X_n$ .                                                                                                                          |
| $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$                        | Observed sample mean for data $x_1, \dots, x_n$ .                                                                                                                                          |
| $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$            | Sample variance as a statistic.                                                                                                                                                            |
| $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$            | Observed sample variance.                                                                                                                                                                  |
| $S = \sqrt{S^2}$ , $s = \sqrt{s^2}$                             | Sample standard deviation as a statistic and its observed value.                                                                                                                           |
| $\hat{\theta}$                                                  | Estimator or estimate of a parameter $\theta$ , according to context.                                                                                                                      |
| $\hat{p}$                                                       | Estimator or estimate of a probability parameter $p$ ; commonly the sample proportion.                                                                                                     |
| $\text{SE}(\hat{\theta})$                                       | Standard error of an estimator $\hat{\theta}$ .                                                                                                                                            |
| $L(\theta; \mathbf{x})$                                         | Likelihood function for parameter $\theta$ given observed data $\mathbf{x}$ .                                                                                                              |
| $\ell(\theta; \mathbf{x}) = \ln L(\theta; \mathbf{x})$          | Log-likelihood function.                                                                                                                                                                   |
| $H_0$                                                           | Null hypothesis.                                                                                                                                                                           |
| $H_1$ , $H_a$                                                   | Alternative hypothesis; $H_1$ is preferred.                                                                                                                                                |
| $\alpha$                                                        | Significance level; an upper bound on the Type I error probability under the specified test.                                                                                               |
| $\beta$                                                         | Type II error probability under a specified alternative.                                                                                                                                   |
| $\nu$                                                           | Degrees of freedom.                                                                                                                                                                        |
| $X \sim \text{Bern}(p)$                                         | Bernoulli distribution with success probability $p$ .                                                                                                                                      |
| $X \sim \text{Bin}(n, p)$                                       | Binomial distribution.                                                                                                                                                                     |
| $X \sim \text{Geom}(p)$ ,<br>$\mathbb{P}(X = k) = (1-p)^{k-1}p$ | Geometric distribution on $\mathbb{N}$ , counting trials up to and including the first success.                                                                                            |
| $X \sim \text{Pois}(\lambda)$                                   | Poisson distribution with parameter $\lambda$ .                                                                                                                                            |
| $X \sim \text{Unif}(a, b)$                                      | Continuous uniform distribution on $[a, b]$ .                                                                                                                                              |
| $X \sim \text{Exp}(\lambda)$                                    | Exponential distribution with rate $\lambda$ .                                                                                                                                             |
| $X \sim \Gamma(\alpha, \beta)$                                  | Gamma distribution with shape $\alpha > 0$ and rate $\beta > 0$ ;<br>$f_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$ for $x > 0$ .                                |
| $X \sim \Gamma(\alpha, \theta)$                                 | Gamma distribution with shape $\alpha > 0$ and scale $\theta > 0$ ;<br>$f_X(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}$ for $x > 0$ , with $\theta = 1/\beta$ . |

| Notation                                                          | Meaning                                                                             |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| $X \sim \mathcal{N}(\mu, \sigma^2),$<br>$X \sim N(\mu, \sigma^2)$ | Normal distribution with mean $\mu$ and variance $\sigma^2$ .                       |
| $X \sim \chi_\nu^2$                                               | Chi-squared distribution with $\nu$ degrees of freedom.                             |
| $X \sim t_\nu$                                                    | Student's $t$ distribution with $\nu$ degrees of freedom.                           |
| $X \sim F_{\nu_1, \nu_2}$                                         | $F$ distribution with numerator and denominator degrees of freedom $\nu_1, \nu_2$ . |
| $X_n \xrightarrow{\text{a.s.}} X$                                 | Almost-sure convergence.                                                            |
| $X_n \xrightarrow{\mathbb{P}} X$                                  | Convergence in probability.                                                         |
| $X_n \xrightarrow{L^p} X$                                         | Convergence in $L^p$ .                                                              |
| $X_n \xrightarrow{d} X, X_n \Rightarrow X$                        | Convergence in distribution.                                                        |

## 9 Geometry

In triangle notation,  $A, B, C$  denote vertices and  $a, b, c$  the opposite side lengths. Lowercase letters such as  $\ell, m, n$  may denote lines. The notation  $\overrightarrow{AB}$  may denote either the ray from  $A$  through  $B$  or the vector from  $A$  to  $B$ , according to context. Greek letters such as  $\omega, \Omega, \Gamma$  may be used as local names for circles. In differential geometry, the Frenet frame follows the standard uppercase convention  $\mathbf{T}, \mathbf{N}, \mathbf{B}$ , typeset in bold to emphasize that these are vectors; uppercase italic letters such as  $X, Y$  may denote vector fields in the usual field-specific convention.

| Notation                                        | Meaning                                                                                                                                             |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $\triangle ABC$                                 | Triangle with vertices $A, B, C$ .                                                                                                                  |
| $\angle ABC$                                    | Geometric angle with vertex $B$ .                                                                                                                   |
| $m\angle ABC$                                   | Measure of the angle $\angle ABC$ .                                                                                                                 |
| $\angle(\ell, m)$                               | Directed angle from the line $\ell$ to the line $m$ , taken modulo $\pi$ , when directed angles are being used.                                     |
| $\overline{AB}$                                 | Segment joining $A$ and $B$ .                                                                                                                       |
| $AB$                                            | Length of $\overline{AB}$ .                                                                                                                         |
| $\overrightarrow{AB}$                           | Ray from $A$ through $B$ , or vector from $A$ to $B$ , according to context.                                                                        |
| $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ | Vector from $A$ to $B$ when $\mathbf{a}$ and $\mathbf{b}$ are the corresponding position vectors.                                                   |
| $\overleftrightarrow{AB}$                       | Line through $A$ and $B$ .                                                                                                                          |
| $\ell \parallel m$                              | Lines $\ell$ and $m$ are parallel.                                                                                                                  |
| $\ell \perp m$                                  | Lines $\ell$ and $m$ are perpendicular.                                                                                                             |
| $\triangle ABC \cong \triangle DEF$             | Congruent triangles; corresponding vertices are listed in the same order.                                                                           |
| $\triangle ABC \sim \triangle DEF$              | Similar triangles; corresponding vertices are listed in the same order.                                                                             |
| $[ABC]$                                         | Area of $\triangle ABC$ .                                                                                                                           |
| $C(O, r)$                                       | Circle with center $O$ and radius $r$ .                                                                                                             |
| $\odot O$                                       | Circle centered at $O$ , especially in elementary geometry when the radius is understood.                                                           |
| $(ABC)$                                         | Circle through the noncollinear points $A, B, C$ ; in particular, the circumcircle of $\triangle ABC$ , in elementary or olympiad geometry context. |
| $\widehat{AB}, \widehat{ACB}$                   | Arcs specified by their endpoints, with an additional point used when needed to distinguish the intended arc.                                       |
| $O, I, H, G$                                    | Circumcenter, incenter, orthocenter, and centroid of a triangle, respectively.                                                                      |

| Notation                                                            | Meaning                                                                                                                                                               |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $R, r$                                                              | Circumradius and inradius of a triangle, respectively.                                                                                                                |
| $s = \frac{a+b+c}{2}$                                               | Semiperimeter of a triangle with side lengths $a, b, c$ .                                                                                                             |
| $d(P, Q)$                                                           | Distance between the points $P$ and $Q$ .                                                                                                                             |
| $\text{dist}(P, A)$                                                 | Distance from the point $P$ to the set $A$ .                                                                                                                          |
| $\text{length}(C), \ell(C)$                                         | Length of the curve $C$ .                                                                                                                                             |
| $\text{area}(S)$                                                    | Area of the surface $S$ .                                                                                                                                             |
| $\text{vol}(\Omega)$                                                | Volume of the region $\Omega$ .                                                                                                                                       |
| $\text{diam}(A)$                                                    | Diameter of the set $A$ .                                                                                                                                             |
| $\text{conv}(A)$                                                    | Convex hull of the set $A$ .                                                                                                                                          |
| $\text{aff}(A)$                                                     | Affine hull of the set $A$ .                                                                                                                                          |
| $\mathbf{x} = \mathbf{p} + t\mathbf{v}$                             | Parametric equation of a line through $\mathbf{p}$ in the direction $\mathbf{v}$ .                                                                                    |
| $\mathbf{n} \cdot (\mathbf{x} - \mathbf{p}) = 0$                    | Hyperplane through $\mathbf{p}$ with normal vector $\mathbf{n}$ .                                                                                                     |
| $\mathbb{R}^n$                                                      | $n$ -dimensional real Euclidean space.                                                                                                                                |
| $S^n = \{\mathbf{x} \in \mathbb{R}^{n+1} \mid \ \mathbf{x}\  = 1\}$ | Unit $n$ -sphere.                                                                                                                                                     |
| $S_r^{n-1}(\mathbf{a})$                                             | Sphere of radius $r$ centered at $\mathbf{a} \in \mathbb{R}^n$ .                                                                                                      |
| $\text{Isom}(X)$                                                    | Isometry group of the metric or geometric space $X$ .                                                                                                                 |
| $\mathbb{RP}^n$                                                     | Real projective $n$ -space.                                                                                                                                           |
| $\mathbb{CP}^n$                                                     | Complex projective $n$ -space.                                                                                                                                        |
| $[x_0 : \cdots : x_n]$                                              | Homogeneous coordinates in projective space.                                                                                                                          |
| $\mathbb{H}^n$                                                      | $n$ -dimensional hyperbolic space.                                                                                                                                    |
| $d_{\mathbb{H}}$                                                    | Hyperbolic distance, when the hyperbolic metric is understood.                                                                                                        |
| $\gamma: I \rightarrow \mathbb{R}^n$                                | Parametrized curve, where $I \subseteq \mathbb{R}$ is an interval.                                                                                                    |
| $\dot{\gamma}(t)$                                                   | Velocity or tangent vector of a differentiable parametrized curve; for a curve in a manifold $M$ , $\dot{\gamma}(t) \in T_{\gamma(t)}M$ .                             |
| $s$                                                                 | Arc-length parameter, when so specified.                                                                                                                              |
| $\mathbf{T}, \mathbf{N}, \mathbf{B}$                                | Unit tangent, principal normal, and binormal vectors of the Frenet frame.                                                                                             |
| $\kappa$                                                            | Curvature of a regular curve.                                                                                                                                         |
| $\tau$                                                              | Torsion of a space curve.                                                                                                                                             |
| $T_p M$                                                             | Tangent space of the manifold or regular surface $M$ at $p$ .                                                                                                         |
| $\mathbf{n}: M \rightarrow S^2$                                     | Gauss map of an oriented surface $M \subseteq \mathbb{R}^3$ , assigning its unit normal vector.                                                                       |
| $\text{I}, \text{II}$                                               | First and second fundamental forms of a regular surface.                                                                                                              |
| $\kappa_1, \kappa_2$                                                | Principal curvatures of a regular surface.                                                                                                                            |
| $K = \kappa_1 \kappa_2$                                             | Gaussian curvature of a regular surface.                                                                                                                              |
| $H = \frac{\kappa_1 + \kappa_2}{2}$                                 | Mean curvature of a regular surface under the average convention.                                                                                                     |
| $(M, g)$                                                            | Riemannian manifold $M$ with Riemannian metric $g$ .                                                                                                                  |
| $g_p$                                                               | Riemannian inner product on $T_p M$ .                                                                                                                                 |
| $\nabla_X Y$                                                        | Covariant derivative of the vector field $Y$ in the direction of the vector field $X$ ; in Riemannian geometry, $\nabla$ commonly denotes the Levi-Civita connection. |
| $\Gamma_{ij}^k$                                                     | Christoffel symbols of a connection in local coordinates.                                                                                                             |
| $\exp_p: \mathcal{D}_p \rightarrow M$                               | Riemannian exponential map at $p$ , where $\mathcal{D}_p \subseteq T_p M$ is its maximal domain. If $M$ is geodesically complete, then $\mathcal{D}_p = T_p M$ .      |
| $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$                            | Geodesic equation for an affinely parametrized curve $\gamma$ .                                                                                                       |

## 10 Number Theory

| Notation                                                  | Meaning                                                                                       |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| $a \mid b$                                                | $a$ divides $b$ .                                                                             |
| $a \nmid b$                                               | $a$ does not divide $b$ .                                                                     |
| $\gcd(a, b)$                                              | Greatest common divisor, taken nonnegative.                                                   |
| $\text{lcm}(a, b)$                                        | Least common multiple.                                                                        |
| $\gcd(a_1, \dots, a_n),$<br>$\text{lcm}(a_1, \dots, a_n)$ | Greatest common divisor and least common multiple of finitely many integers.                  |
| $p^k \parallel n$                                         | $p^k$ exactly divides $n$ : $p^k \mid n$ and $p^{k+1} \nmid n$ ; equivalently, $v_p(n) = k$ . |
| $v_p(n), \nu_p(n)$                                        | $p$ -adic valuation of $n$ ; $v_p(n)$ is preferred.                                           |
| $a \equiv b \pmod{n}$                                     | $a$ is congruent to $b$ modulo $n$ .                                                          |
| $[a]_n$                                                   | Residue class of $a$ modulo $n$ .                                                             |
| $\mathbb{Z}/n\mathbb{Z}$                                  | Ring of residue classes modulo $n$ .                                                          |
| $(\mathbb{Z}/n\mathbb{Z})^\times$                         | Multiplicative group of units modulo $n$ .                                                    |
| $a^{-1} \pmod{n}$                                         | Multiplicative inverse of $a$ modulo $n$ .                                                    |
| $\text{ord}_n(a)$                                         | Multiplicative order of $a$ modulo $n$ .                                                      |
| $\mathbb{F}_q, \text{GF}(q)$                              | Finite field with $q$ elements, where $q$ is a prime power; $\mathbb{F}_q$ is preferred.      |
| $\left(\frac{a}{p}\right)$                                | Legendre symbol, for an odd prime $p$ .                                                       |
| $\left(\frac{a}{n}\right)$                                | Jacobi symbol, for an odd positive integer $n$ .                                              |
| $\varphi(n)$                                              | Euler totient function.                                                                       |
| $\sum_{d \mid n}$                                         | Sum over the positive divisors $d$ of $n$ .                                                   |
| $\prod_{p \mid n}$                                        | Product over the prime divisors $p$ of $n$ .                                                  |
| $\sigma_k(n) = \sum_{d \mid n} d^k$                       | Sum of the $k$ th powers of the positive divisors of $n$ .                                    |
| $\tau(n) = \sigma_0(n)$                                   | Number of positive divisors of $n$ .                                                          |
| $\sigma(n) = \sigma_1(n)$                                 | Sum of positive divisors of $n$ .                                                             |
| $\mu(n)$                                                  | Möbius function.                                                                              |
| $\omega(n)$                                               | Number of distinct prime divisors of $n$ .                                                    |
| $\Omega(n)$                                               | Number of prime factors of $n$ , counted with multiplicity.                                   |
| $\text{rad}(n)$                                           | Product of the distinct prime divisors of $n$ .                                               |
| $(f * g)(n) = \sum_{d \mid n} f(d)g(n/d)$                 | Dirichlet convolution of arithmetic functions $f$ and $g$ .                                   |
| $\pi(x)$                                                  | Number of primes not exceeding $x$ .                                                          |
| $[a_0; a_1, a_2, \dots]$                                  | Simple continued fraction.                                                                    |
| $[a_0; a_1, \dots, a_n] = \frac{p_n}{q_n}$                | $n$ th convergent of a continued fraction.                                                    |
| $\Phi_n(x)$                                               | $n$ th cyclotomic polynomial.                                                                 |
| $\mathbb{Z}[i]$                                           | Ring of Gaussian integers.                                                                    |
| $K = \mathbb{Q}(\sqrt{d})$                                | Quadratic field, with squarefree $d \in \mathbb{Z}$ and $d \neq 1$ .                          |
| $\mathcal{O}_K$                                           | Ring of integers of the number field $K$ .                                                    |
| $N(\mathfrak{a}) =  \mathcal{O}_K/\mathfrak{a} $          | Absolute norm of a nonzero integral ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ .            |
| $D_K, \text{disc}(K)$                                     | Discriminant of the number field $K$ ; $D_K$ is preferred.                                    |
| $\text{Cl}(K)$                                            | Ideal class group of the number field $K$ .                                                   |
| $h_K =  \text{Cl}(K) $                                    | Class number of the number field $K$ .                                                        |

| Notation | Meaning |
|----------|---------|
|----------|---------|

## 11 Algebra

$G, H$  usually denote groups,  $R, S$  rings,  $I, J$  ideals,  $M, N$  modules, and  $F, K, L$  fields. Rings are associative and have an identity unless otherwise stated; commutativity is not assumed. Ring homomorphisms preserve multiplicative identities. Unless otherwise stated, ideals are two-sided. The zero ring is allowed, while fields and division rings are nonzero. Modules are left modules unless otherwise stated.

| Notation                   | Meaning                                                                                               |
|----------------------------|-------------------------------------------------------------------------------------------------------|
| $G$                        | Group.                                                                                                |
| $e, 1_G$                   | Identity element of a group.                                                                          |
| $g^{-1}$                   | Inverse of $g \in G$ .                                                                                |
| $ G $                      | Order (cardinality) of $G$ .                                                                          |
| $\text{ord}(g),  g $       | Order of the element $g$ .                                                                            |
| $\langle S \rangle$        | Subgroup generated by $S \subseteq G$ .                                                               |
| $\langle g \rangle$        | Cyclic subgroup generated by $g$ .                                                                    |
| $H \leq G$                 | $H$ is a subgroup of $G$ .                                                                            |
| $H \trianglelefteq G$      | $H$ is a normal subgroup of $G$ .                                                                     |
| $gH, Hg$                   | Left and right cosets of $H$ in $G$ .                                                                 |
| $[G : H]$                  | Index of $H$ in $G$ .                                                                                 |
| $G/H$                      | Quotient group, for $H \trianglelefteq G$ .                                                           |
| $\ker \varphi$             | Kernel of the homomorphism $\varphi$ .                                                                |
| $\text{im } \varphi$       | Image of the homomorphism $\varphi$ .                                                                 |
| $\text{Aut}(G)$            | Automorphism group of $G$ .                                                                           |
| $\text{Inn}(G)$            | Inner automorphism group of $G$ .                                                                     |
| $Z(G)$                     | Center of $G$ .                                                                                       |
| $C_G(A), C_G(x)$           | Centralizer in $G$ of $A \subseteq G$ or $x \in G$ .                                                  |
| $N_G(A), N_G(H)$           | Normalizer in $G$ of $A \subseteq G$ or $H \leq G$ .                                                  |
| $[x, y] := x^{-1}y^{-1}xy$ | Commutator of $x$ and $y$ .                                                                           |
| $[G, G], G'$               | Commutator (derived) subgroup of $G$ .                                                                |
| $\text{Syl}_p(G)$          | Set of Sylow $p$ -subgroups of $G$ .                                                                  |
| $n_p$                      | Number of Sylow $p$ -subgroups of $G$ .                                                               |
| $G \curvearrowright X$     | Action of $G$ on $X$ .                                                                                |
| $g \cdot x$                | Action of $g \in G$ on $x \in X$ .                                                                    |
| $G \cdot x$                | Orbit of $x$ under the action of $G$ .                                                                |
| $\text{Stab}_G(x), G_x$    | Stabilizer of $x$ in $G$ .                                                                            |
| $G \times H$               | Direct product of groups.                                                                             |
| $N \rtimes_{\varphi} H$    | Semidirect product determined by $\varphi: H \rightarrow \text{Aut}(N)$ , with $N$ the normal factor. |
| $\langle S \mid R \rangle$ | Group presentation with generators $S$ and relations $R$ .                                            |
| $C_n$                      | Cyclic group of order $n$ .                                                                           |
| $D_n$                      | Dihedral group of symmetries of a regular $n$ -gon; $ D_n  = 2n$ .                                    |
| $S_n, \mathfrak{S}_n$      | Symmetric group on $n$ symbols.                                                                       |
| $A_n, \mathfrak{A}_n$      | Alternating group on $n$ symbols.                                                                     |
| $V_4$                      | Klein four-group.                                                                                     |
| $Q_8$                      | Quaternion group of order 8.                                                                          |
| $\text{GL}_n(F)$           | General linear group over $F$ .                                                                       |

| Notation                                  | Meaning                                                                                                                                                                                                                                        |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathrm{SL}_n(F)$                        | Special linear group over $F$ .                                                                                                                                                                                                                |
| $R$                                       | Ring.                                                                                                                                                                                                                                          |
| $0_R, 1_R$                                | Additive and multiplicative identities of $R$ , when disambiguation is useful.                                                                                                                                                                 |
| $R^\times$                                | Group of units of $R$ .                                                                                                                                                                                                                        |
| $I, J$                                    | Two-sided ideals of $R$ , unless otherwise stated.                                                                                                                                                                                             |
| $\mathfrak{p}, \mathfrak{q}$              | Prime ideals.                                                                                                                                                                                                                                  |
| $\mathfrak{m}, \mathfrak{n}$              | Maximal ideals.                                                                                                                                                                                                                                |
| $(a)$                                     | Two-sided ideal generated by $a$ ; in a commutative ring, the usual principal ideal.                                                                                                                                                           |
| $(a_1, \dots, a_n)$                       | Two-sided ideal generated by $a_1, \dots, a_n$ .                                                                                                                                                                                               |
| $I + J$                                   | Sum of the ideals $I$ and $J$ .                                                                                                                                                                                                                |
| $IJ$                                      | Product of the ideals $I$ and $J$ .                                                                                                                                                                                                            |
| $R/I$                                     | Quotient ring.                                                                                                                                                                                                                                 |
| $R[x], R[x_1, \dots, x_n]$                | Polynomial rings over $R$ in commuting central indeterminates; the indeterminates commute with the coefficients and with one another. Noncommutative polynomial constructions are specified separately.                                        |
| $\deg f$                                  | Degree of the polynomial $f$ .                                                                                                                                                                                                                 |
| $M_n(R)$                                  | Ring of $n \times n$ matrices over $R$ .                                                                                                                                                                                                       |
| $R[G]$                                    | Group ring of $G$ over $R$ .                                                                                                                                                                                                                   |
| $\mathrm{Frac}(R)$                        | Fraction field of the integral domain $R$ .                                                                                                                                                                                                    |
| $\mathrm{char} R$                         | Characteristic of $R$ .                                                                                                                                                                                                                        |
| $\mathbb{H}$                              | Quaternion division algebra over $\mathbb{R}$ .                                                                                                                                                                                                |
| $M, N$                                    | Left $R$ -modules, unless otherwise stated.                                                                                                                                                                                                    |
| ${}_R M, M_R$                             | Left and right $R$ -modules, when the side is emphasized.                                                                                                                                                                                      |
| $M/N$                                     | Quotient module, for a submodule $N$ of $M$ .                                                                                                                                                                                                  |
| $M \oplus N$                              | Direct sum of modules.                                                                                                                                                                                                                         |
| $\bigoplus_{i \in \Lambda} M_i$           | Direct sum of an indexed family of modules, with index set $\Lambda$ .                                                                                                                                                                         |
| $R^n$                                     | Free $R$ -module of rank $n$ .                                                                                                                                                                                                                 |
| $\mathrm{Hom}_R(M, N)$                    | Set of $R$ -module homomorphisms from $M$ to $N$ .                                                                                                                                                                                             |
| $\mathrm{End}_R(M)$                       | Endomorphism ring of the $R$ -module $M$ .                                                                                                                                                                                                     |
| $\mathrm{Aut}_R(M)$                       | Automorphism group of the $R$ -module $M$ .                                                                                                                                                                                                    |
| $\mathrm{Ann}_R(M)$                       | Annihilator of the $R$ -module $M$ .                                                                                                                                                                                                           |
| $M_R \otimes_R {}_R N, M \otimes_R N$     | Tensor product over $R$ ; the first notation emphasizes that $M$ is a right $R$ -module and $N$ a left $R$ -module. The side distinction is normally suppressed when $R$ is commutative or the required module structures are otherwise clear. |
| $K/F$                                     | Field extension of $F$ by $K$ .                                                                                                                                                                                                                |
| $[K : F]$                                 | Degree of the field extension $K/F$ .                                                                                                                                                                                                          |
| $F(\alpha), F(\alpha_1, \dots, \alpha_n)$ | Fields generated over $F$ by the indicated elements.                                                                                                                                                                                           |
| $F[\alpha]$                               | Subring (equivalently, $F$ -subalgebra) generated by $F$ and $\alpha$ .                                                                                                                                                                        |
| $m_{\alpha, F}(x)$                        | Minimal polynomial of $\alpha$ over $F$ .                                                                                                                                                                                                      |
| $\overline{F}$                            | Algebraic closure of $F$ .                                                                                                                                                                                                                     |
| $\mathrm{Gal}(L/K)$                       | Galois group of $L/K$ .                                                                                                                                                                                                                        |
| $L^H$                                     | Fixed field of $H \leq \mathrm{Gal}(L/K)$ .                                                                                                                                                                                                    |
| $\mathrm{Tr}_{L/K}(\alpha)$               | Field trace of $\alpha \in L$ from $L$ to $K$ .                                                                                                                                                                                                |
| $N_{L/K}(\alpha)$                         | Field norm of $\alpha \in L$ from $L$ to $K$ .                                                                                                                                                                                                 |

| Notation           | Meaning                                |
|--------------------|----------------------------------------|
| $\text{trdeg}_K L$ | Transcendence degree of $L$ over $K$ . |

## 12 Topology

| Notation                                                           | Meaning                                                                                                                         |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| $(X, \mathcal{T})$                                                 | Topological space $X$ with topology $\mathcal{T}$ .                                                                             |
| $\mathcal{T}$                                                      | Topology on the underlying set $X$ , when $X$ is understood.                                                                    |
| $\mathcal{B}$                                                      | Basis for a topology.                                                                                                           |
| $\mathcal{S}$                                                      | Subbasis for a topology.                                                                                                        |
| $A^\circ$ , $\text{int } A$                                        | Interior of $A$ .                                                                                                               |
| $\overline{A}$ , $\text{cl } A$                                    | Closure of $A$ .                                                                                                                |
| $\partial A$                                                       | Boundary of $A$ .                                                                                                               |
| $(X, d)$                                                           | Metric space $X$ with metric $d$ .                                                                                              |
| $\mathcal{T}_A = \{A \cap U \mid U \in \mathcal{T}\}$              | Subspace topology on $A \subseteq X$ .                                                                                          |
| $\pi_i: \prod_{j \in J} X_j \rightarrow X_i$                       | Coordinate projection of a product space, for $i \in J$ .                                                                       |
| $X/\sim$                                                           | Quotient space when the quotient set is equipped with the quotient topology.                                                    |
| $q: X \rightarrow X/\sim$                                          | Canonical quotient map.                                                                                                         |
| $X/A$                                                              | Quotient space obtained by collapsing the subspace $A \subseteq X$ to a point.                                                  |
| $C(X, Y)$                                                          | Set of continuous maps from $X$ to $Y$ .                                                                                        |
| $X \cong Y$                                                        | Homeomorphic spaces.                                                                                                            |
| $\text{Homeo}(X)$                                                  | Group of self-homeomorphisms of $X$ .                                                                                           |
| $\iota: A \hookrightarrow X$                                       | Inclusion map of a subspace $A \subseteq X$ .                                                                                   |
| $f: X \hookrightarrow Y$                                           | Topological embedding when explicitly stated; the hook arrow by itself retains its general meaning of injectivity.              |
| $(X, x_0)$                                                         | Pointed space with basepoint $x_0$ .                                                                                            |
| $I = [0, 1]$                                                       | Unit interval, especially for paths and homotopies.                                                                             |
| $\gamma: I \rightarrow X$                                          | Path in $X$ .                                                                                                                   |
| $\gamma(0) = \gamma(1) = x_0$                                      | Loop based at $x_0$ .                                                                                                           |
| $H: X \times I \rightarrow Y$                                      | Homotopy between maps $X \rightarrow Y$ .                                                                                       |
| $f \simeq g$                                                       | Homotopic maps $f, g: X \rightarrow Y$ .                                                                                        |
| $X \simeq Y$                                                       | Homotopy equivalent spaces.                                                                                                     |
| $[\gamma]$                                                         | Homotopy class of a path or loop $\gamma$ , according to context.                                                               |
| $\pi_1(X, x_0)$                                                    | Fundamental group of $X$ based at $x_0$ .                                                                                       |
| $\pi_n(X, x_0)$                                                    | $n$ th homotopy group of the pointed space $(X, x_0)$ , for $n \geq 1$ .                                                        |
| $f_*$                                                              | Homomorphism induced by a map $f$ on a covariant invariant such as homotopy or homology, with the invariant clear from context. |
| $f^*$                                                              | Homomorphism induced by a map $f$ on a contravariant invariant such as cohomology, with the invariant clear from context.       |
| $p: \tilde{X} \rightarrow X$                                       | Covering map, with $\tilde{X}$ a covering space of $X$ .                                                                        |
| $\tilde{X}$                                                        | Covering space; it denotes a universal cover only when explicitly specified.                                                    |
| $\text{Deck}(p)$                                                   | Group of deck transformations of the covering $p$ .                                                                             |
| $D^n = \{\mathbf{x} \in \mathbb{R}^n \mid \ \mathbf{x}\  \leq 1\}$ | Closed $n$ -disk.                                                                                                               |
| $X \vee Y$                                                         | Wedge sum of pointed spaces $X$ and $Y$ .                                                                                       |

| Notation                  | Meaning                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| $T_0, T_1, T_2, T_3, T_4$ | Separation axioms; here $T_2$ means Hausdorff, $T_3$ means $T_1$ and regular, and $T_4$ means $T_1$ and normal.  |
| $M^n$                     | $n$ -dimensional topological manifold, when the dimension is displayed.                                          |
| $\partial M$              | Boundary of a manifold $M$ .                                                                                     |
| $\Sigma_g$                | Closed connected orientable surface of genus $g$ .                                                               |
| $M \# N$                  | Connected sum of manifolds $M$ and $N$ .                                                                         |
| $\chi(X)$                 | Euler characteristic of $X$ .                                                                                    |
| $H_n(X; G)$               | $n$ th homology group of $X$ with coefficients in $G$ .                                                          |
| $H_n(X)$                  | Integral homology group $H_n(X; \mathbb{Z})$ when coefficients are omitted.                                      |
| $\tilde{H}_n(X; G)$       | $n$ th reduced homology group of $X$ with coefficients in $G$ .                                                  |
| $H^n(X; G)$               | $n$ th cohomology group of $X$ with coefficients in $G$ .                                                        |
| $b_n(X)$                  | $n$ th Betti number; for example, $b_n(X) = \dim_{\mathbb{Q}} H_n(X; \mathbb{Q})$ when this dimension is finite. |

### 13 Miscellaneous Notation

| Notation                       | Meaning                                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $\lfloor x \rfloor$            | Floor of $x$ .                                                                                                                                      |
| $\lceil x \rceil$              | Ceiling of $x$ .                                                                                                                                    |
| $\{x\}$                        | Fractional part $x - \lfloor x \rfloor$ , when unambiguous.                                                                                         |
| $\operatorname{sgn}(x)$        | Sign of a real number or permutation, according to context.                                                                                         |
| $a \propto b$                  | $a$ is proportional to $b$ ; in a functional context, the proportionality factor is independent of the variable under discussion.                   |
| $O(g(n))$                      | Big-O upper asymptotic bound.                                                                                                                       |
| $o(g(n))$                      | Little-o strict asymptotic bound.                                                                                                                   |
| $\Theta(g(n))$                 | Two-sided asymptotic order.                                                                                                                         |
| $\Omega(g(n))$                 | Asymptotic lower bound.                                                                                                                             |
| $\sum_{\text{cyc}} f(a, b, c)$ | Cyclic sum over the cyclic permutations of the displayed variables; for three variables, $f(a, b, c) + f(b, c, a) + f(c, a, b)$ .                   |
| $\sum_{\text{sym}} f(a, b, c)$ | Symmetric sum over the distinct permutations of the displayed variables in the summand; the variables and any normalization are stated when needed. |
| $\Delta q$                     | Finite change in a quantity $q$ , especially in a discrete or applied context.                                                                      |
| $\delta_{ij}$                  | Kronecker delta: 1 if $i = j$ and 0 otherwise.                                                                                                      |
| $\epsilon_{ijk}$               | Three-dimensional Levi-Civita symbol: the sign of the permutation $(i, j, k)$ when $(i, j, k)$ is a permutation of $(1, 2, 3)$ , and 0 otherwise.   |